The Interview-Compatible Recommender: A Measuring Instrument for Cold-Start Conversational Recommendation

Thesis Chapter A (v2)

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Abstract

Before one can study *how* to interview a cold-start user—which questions to ask, in what order—one needs a recommender that can *be* interviewed: a model that folds an arbitrary-length, mixed-channel set of answers into its belief and re-ranks a large catalogue, without per-user retraining and without trading away full-profile accuracy. We argue that this recommender is a *measuring instrument* for the entire elicitation-policy literature, and that its requirements are more demanding than any single published system meets. The demanding part is *ingestion*: folding an arbitrary-length set of mixed-type answers into a recommender that is already at the accuracy frontier. We state that as five requirements—**R1** state-of-the-art full-profile accuracy, **R2** an open set-valued input interface (any number of observations, and directions that need not have existed when the model was fitted—concepts, entities, phrases, policy-emitted vectors), **R3** a queryable belief over the user, **R4** monotonicity (a truthful answer never lowers ranking quality), and **R5** answers that carry sign and intensity—and an acceptance battery (one verification item per requirement, **V1–V5**, plus two integrity items) that operationalises them. R2 carries most of the weight: R3 is what the instrument *provides* to a policy, and what a policy should do with it is the companion chapter's subject. A survey of six model families shows the literature is *bifurcated*: the models that hit the full-profile bar (shallow linear/autoencoder recommenders such as EASE [40] and RecVAE [38]) are item-only point estimators, while the models with an uncertainty-native, multi-channel belief (soft-attribute Gaussian elicitation [5], unified item+attribute bandits [23], conjugate critique folds [44]) do not report full-profile accuracy at the R1 bar. We are aware of no published system that occupies the conjunction of all five. The instrument that closes this gap is a *frozen* certified shallow-SOTA tower over whose latent every channel is folded as a direction paired with a value, with an analytic conjugate-Gaussian belief maintained over the same latent for a policy to query; three of its input legs—signed/graded values, concepts whose vocabulary is not fixed when the model is fitted, and continuous direction tokens—are architecturally inexpressible in the item-indicator basis of the R1-bar models. Each requirement carries one verification item, V1–V5, and all five are met. This chapter contributes the requirement framework, the taxonomy and gap analysis, the instrument's design, and the experimental protocol (a metric-certification bridge to the published ML-20M numbers, then V1–V5 on the ML-25M harness). Our in-house RecVAE realisation sets the bar at full/tail NDCG@10 0.3540/0.2497 on the canonical ML-25M protocol, and the graded-native interview tower now *meets* it: a single certified training run reaches 0.3482/0.2462 on the held-out test cohort—a paired-bootstrap tie with the bar—while reproducing the frozen decoder's ranking exactly at the empty interview, so V1 (full strength preserved) is passed rather than assumed. The remaining verification items have all been run and are met. With the tower fixed, a quantity the elicitation literature has long treated qualitatively—how often a cold user can answer the question at all—becomes directly measurable: in our harness, at an eight-question budget,

a user answers 2.18 of the item questions and 6.02 of the concept questions, and the concept channel leads on the opening of an interview and on tail accuracy throughout.

Contents

1 Introduction: the recommender as a measuring instrument

A *cold-start* conversational recommender must elicit taste from a short interview and then rank a large catalogue. Two questions are entangled: *what to ask* (the elicitation policy) and *how to turn answers into recommendations* (the recommender). The elicitation-policy literature is large and active, but almost every comparison in it is confounded: a policy's apparent value depends on the recommender it drives, and reported gains routinely conflate a better *question-selection strategy* with a better *answer-ingestion model*. Before we can measure elicitation strategies in isolation, we need to fix the recommender—and not just any recommender, but one strong and flexible enough that a good interview is not wasted on it. We call such a recommender an *interview-compatible instrument*.

Five requirements. An instrument fit for measuring conversational elicitation must simultaneously satisfy:

R1 — full-profile SOTA. On a full interaction history it must rank at the level of the strongest shallow collaborative-filtering baselines (EASE [40], RecVAE [38]), so that elicitation is measured against a recommender nobody can dismiss as weak [12].

R2 — an open set-valued input interface. It must ingest an evidence set of *any* cardinality, from 0 (fully cold) upward, without per-user retraining—and, the part that bites, the members of that set must be *arbitrary embeddings rather than pre-committed ones.* Any-length input alone is undemanding: every recommender folds an arbitrary number of *its own* columns. The requirement is that a vector the model never saw when it was fitted—an out-of-catalogue concept, an entity, a phrase, a direction a policy emits—is a legal member of the set. That demands both a set-structured input and a latent space arbitrary vectors can be inserted into, which is an architectural property and not a tuning one.

Why this is an elicitation requirement and not an engineering preference. A cold interview is rate-limited by what the user can actually answer, and answerability rises with abstraction: a question about a kind of film presupposes nothing, where a question about a particular film presupposes having seen it (§6.4). An instrument restricted to catalogue items is therefore capped by the user's viewing history; one restricted to a curated attribute list is capped by whoever wrote the list. Neither cap is a property of the user. Since abstraction has no natural ceiling—"psychological thrillers", "films about grief", "the quiet dignity of losing"—any vocabulary fixed at training time forecloses part of what a user could have told us, and which part is decided before the user arrives.

R3 — queryable belief. It must expose an explicit covariance over the user belief, not only a point estimate, so that a policy *can* act on what is not yet known. The instrument's obligation is to provide and verify that belief; whether and how a policy exploits it is a separate question.

R4 — monotonicity. A truthful answer must never *lower* ranking quality; adding evidence should, in expectation, only help.

R5 — expressive answers. An answer must be able to say what a star rating says and an indicator cannot: that the user *dislikes* something, and *how strongly.* Sign and intensity are information a one-hot cannot carry at all.

The bifurcation observation. The central empirical fact this chapter documents is that R1 and {R2, R3, R4, R5} are *anti-correlated across the published literature.* The models that hit the full-profile bar—closed-form linear autoencoders (EASE [40], EDLAE [41]) and amortised VAEs (Mult-VAE [25], RecVAE [38]), together with the training-free graph filters (GF-CF [37], BSPM [8], Turbo-CF [32]) that claim the frontier on other corpora—are *item-only point estimators* on a one-hot indicator basis. The models with an uncertainty-native, multi-channel belief—soft-attribute Gaussian elicitation [5], item+attribute Thompson bandits [23], conjugate critique folds over a frozen factorisation [44], and per-item Bayesian NL elicitation [3]—do *not* report full-profile accuracy at the R1 bar. We call these the *accuracy corner* and the *belief corner.* No published system we are aware of occupies both (§4).

Contributions. This chapter is the instrument paper of the thesis. It does not claim a new elicitation policy (that is the companion method chapter); it claims the *recommender that makes policy measurement possible.* Concretely:

1. **One latent, many channels, at the accuracy frontier.** A frozen tower that ties the strongest shallow baseline on a shared ruler, over whose latent every channel is folded as a

direction paired with a value: item factors, out-of-catalogue concept directions, and directions produced outside the catalogue entirely. The unification is of the *representation*—one space, one form of observation—not of the code path. We frame R1∧R2∧R3∧R4∧R5 as a single requirement no published system meets, and give a six-family taxonomy and per-near-miss gap analysis (§3, §4) locating which requirement each candidate fails.

2. **Three architecturally-inexpressible legs.** We isolate the three input channels that are impossible (not merely untuned) in the item-indicator basis of every R1-bar model: (i) signed/graded per-entity values, (ii) concepts whose vocabulary is not fixed at training time, and (iii) continuous direction tokens. These are the narrow, defensible novelty legs of the channel-agnostic interface (§4, §5).

3. **The instrument.** A design for a *frozen* certified shallow-SOTA tower with an analytic conjugate-Gaussian precision-accumulator belief layer over its latent, into which every channel folds as a linear observation (§5). The frozen-tower+exact-conjugacy shape is the one mechanism with a principled monotonicity story [44]. Freezing is *not* what separates us from the soft-attribute line—Göpfert et al. [16] and Bıyık et al. [5] both freeze the collaborative model and learn attribute directions separately, and say so explicitly. What separates us is an *open* concept vocabulary carrying no per-concept supervision, at the R1 accuracy bar (§4).

4. **An acceptance battery as a reusable certificate.** We define three headline outcome gates—full-strength preserved (V1), posterior strictly shrinks per question (V3), and cold-start per-question NDCG rises on *full and tail* versus random (V4)—and, because outcome checks alone admit several false positives (a sign-blind fold, a circular answer model, a decorative covariance), pair them with one verification item per requirement, V1–V5, and two integrity items. We propose this as a portable certificate any interview-compatible recommender should pass (§2, §7).

We are careful throughout to distinguish what is *new* (the conjunction; the frozen-certified-tower wedge; the acceptance battery on a SOTA tower) from what is *pre-empted* and must be cited (attributes-as-directions plus a Bayesian fold [5]; unified item+attribute belief [23]; conjugate folds over a frozen factorisation [44]; permutation-invariant set-encoders with information-gain acquisition [30]; value-carrying set-input cold-start recommenders [26]; the non-degradation-of-information principle [15, 6]). "Uncertainty-native recommender" alone is a crowded 2024–25 area [7] and is *not* claimed as novel.

2 Requirements and gates

We now state the requirements and gates formally. Let the catalogue have n items and let the frozen tower expose a d-dimensional latent in which each item i has a factor row $\mathbf{Q}_i \in \mathbb{R}^d$. An *evidence set* after t questions is a set of observations $\mathcal{E}_t = \{(\boldsymbol{\phi}_j, y_j)\}_{j=1}^{t}$, where $\boldsymbol{\phi}_j \in \mathbb{R}^d$ is the latent direction of the queried entity (an item factor, a concept centroid, an entity embedding, or a continuous direction token) and $y_j \in \mathbb{R}$ is the (possibly graded, possibly signed) answer. The instrument maintains a belief $b(\mathcal{E}_t) = (\boldsymbol{\mu}_t, \boldsymbol{\Sigma}_t)$ over the user latent $\mathbf{u} \in \mathbb{R}^d$ and scores item i by a fixed read-out $s_i(\boldsymbol{\mu}_t)$ of the base tower.

Requirements (formal).

R1. With $\mathcal{E}$ equal to a user's full history, $\mathrm{NDCG@10}\big(s(\boldsymbol{\mu}(\mathcal{E}))\big)$ ties the strongest shallow baseline on a shared harness (Def. 1).

R2. $b(\mathcal{E}_t)$ is defined and computable for every $t \geq 0$ with no gradient step at inference (t=0 yields the prior belief), *and* the admissible $\boldsymbol{\phi}_j$ are all of $\mathbb{R}^d$ rather than a finite set fixed when the model was fitted. The update from $\mathcal{E}_t$ to $\mathcal{E}_{t+1}$ depends on the new observation only through the pair $(\boldsymbol{\phi}_{t+1}, y_{t+1}) \in \mathbb{R}^d \times \mathbb{R}$, so any entity with an embedding is a legal member of the evidence set. The requirement is on the *observation*, not on the number of code paths that consume it.

R3. $\boldsymbol{\Sigma}_t$ is an explicit covariance, shrinking over answers and directionally informative, readable by a policy for any direction w as $w^\top \boldsymbol{\Sigma}_t w$.

R4. For a truthful answer y_{t+1}, $\mathbb{E}\big[\mathrm{NDCG@10}(s(\boldsymbol{\mu}_{t+1}))\big] \geq \mathrm{NDCG@10}(s(\boldsymbol{\mu}_t))$. We treat R4 as an *empirical* property to be demonstrated, not a theorem for a learned ranker (§8).

R5. The answer value y_j carries *sign* and *magnitude*: $y_j < 0$ expresses dislike and $|y_j|$ expresses intensity. An indicator basis has neither—a one-hot can record that an item was seen, not that it was disliked, nor how much.

Definition 1 (Ties SOTA)**.** *A recommender* ties *a baseline on a metric if their scores are within the paired per-user bootstrap confidence interval on one shared harness (same users, items, split, metric depth). A number computed on a different metric/split (e.g. NDCG@10 on ML-25M vs. NDCG@100 on the ML-20M strong-generalisation split) is* not *a tie until bridged (§7.2).*

Verification: one check per requirement. Each requirement carries exactly one verification item, V1–V5. Two integrity items, I1–I2, apply across all five and exist because the per-requirement checks admit false positives (below). Table 1 is the compact reference; the definitions follow. Separately, the *ruler* itself is certified by the published-number bridge of §7.2—that is a property of our measuring apparatus, not of the instrument, and it is reported there.

V1 — full strength and identity (R1). Evaluating on the belief mean with the full profile reproduces the frozen tower's full-profile NDCG@10 (full *and* tail)—a paired-bootstrap tie—and with an *empty* evidence set the ranking is the frozen decoder's exactly, not approximately. The identity half is what licenses reading every later curve as evidence about ingestion rather than about a shifted baseline.

V2 — an open set-valued input interface (R2). Two halves, and only the second can fail. The set half is structural: the belief is defined for every $t \geq 0$ with no gradient step, because the fold is a running sum over observations. The *open* half is measured: a concept must be a specific direction rather than a popularity proxy—ranking its member items above matched non-members in the latent, and lifting the cold ranking beyond what the concept's own popularity explains (its top principal component partialled out)—and a direction obtained for a concept *held out of the adapter's fit*, from its name alone, must recover that concept's own ranking far above chance. That last clause is the one no fixed-column model can attempt. We deliberately do *not* require a high member-lift *after* folding: the fold is trained to predict a user's actual targets, and a user who likes film noir does not like every noir film; an operator that pushed all of a concept's members up the ranking would be behaving as a genre filter, which is a failure mode rather than the property to gate on. The ablation of §6.2 measures that trade-off directly—the additive-union operator reaches member AUC 0.820 and *craters* to 0.0991 full NDCG@10, below the no-answer intercept, while the gated fold sits at 0.626 and reaches 0.1759. What we require instead is that the movement be *specific*: folding a like on a concept must move that concept's members differently from folding a dislike on it, and differently again from folding some other answerable concept.

V3 — queryable belief (R3). The posterior covariance shrinks in expectation over answers; the shrinkage is at least twice as large along the queried $\boldsymbol{\phi}_{t+1}$ as elsewhere; and replacing $\boldsymbol{\Sigma}$ with an isotropic $c\mathbf{I}$ *degrades* a downstream number, else the R3 claim is withdrawn. The belief and the ranking are invariant to answer order and to conflicting re-asks.

V4 — monotonicity (R4). Starting cold, per-question full and tail NDCG@10 *increase* to budget on a question set fixed in advance, so the rise comes from answer content rather than from adaptive selection. Judged by the curve at budget rather than every-step significance, with a canary that a single answer from cold already clears the no-answer intercept on both channels.

V5 — expressive answers (R5). *Sign*: negating every answer lowers NDCG with a CI excluding zero. *Intensity*: neutralising values to "indifferent" while freezing membership costs NDCG, quality is monotone in the stated level, and graded input beats a fair binarising ablation where the extra bits should matter most—on the short interview, before collaborative signal fills in.

Integrity items (across all five):

I1 — existential controls. Wrong-user, shuffled, and placebo-constant answers through the identical pipeline *gain nothing* relative to the true-answer arm (a control may legitimately fall *below* the no-answer intercept—a stranger's taste is worse than no information, not equivalent to it), and duplicating an answer changes nothing (the belief responds to information, not cardinality).

I2 — answers carry no recommender geometry. Every answer is computed from observed behaviour alone: an item answer is the user's own rating, and a concept answer is a log watch-lift, $\log_2\big((n_c+\frac{1}{2})/(e_c+\frac{1}{2})\big)$, formed from the user's watched members n_c and the count e_c expected from their volume and the concept's catalogue popularity. Both are read from the user's *fold-in* history; the held-out target half is excluded by construction, so an answer cannot encode its own target. No latent, no factor, no score and no quantity derived from the recommender enters an answer at any point, and the absence is asserted in code rather than assumed. The gain therefore cannot be the instrument recovering its own geometry, because that geometry is nowhere in the input.

Why the outcome checks alone are insufficient. V1, V3 and V4 are outcome checks, and several broken instruments pass all three. A *sign-blind* fold that updates its belief from only the *direction* of the queried entity but never the answer *value* still reproduces the frozen tower (V1), still shrinks $\boldsymbol{\Sigma}$ along $\boldsymbol{\phi}$ per question (V3), and still beats a random-question control—because beating random certifies that a good question was *selected*, not that the answer was *ingested*; only V5's sign-flip and V4's fixed-bank arm expose it. A *circular* answer model that generates answers from the instrument's own latent geometry would inflate the V4 curve with no transferable signal. The risk is not hypothetical in this literature: evaluations commonly grant the simulated user privileged access to the target—PEBOL's simulator answers from the ground-truth item description [3], and unified item-and-attribute CRS work assumes the user holds and reveals clear preferences over every attribute and item [10]. I2 closes this by construction rather than by measurement (§2). And a *decorative* covariance that is maintained but never enters a reported number passes V1 and V5 trivially; only replacing $\boldsymbol{\Sigma}$ with an isotropic matrix and demanding a degradation—V3's third clause—shows whether the uncertainty is load-bearing. V2, V5 and the two integrity items exist precisely to close these outcome-check false positives.

Check	One-line criterion	Failure class it excludes
V1 (R1)	Belief mean reproduces the frozen tower's full/tail NDCG@10 (tie); an empty interview is the decoder exactly.	Full-profile regression from the belief layer.
V2 (R2)	Any $t \geq 0$, no gradient step; *and* a concept is a specific direction, not a popularity proxy, with a held-out concept folding from its name alone.	Per-user retraining; a vocabulary fixed at fit time; popularity-parasite concepts.
V3 (R3)	$\boldsymbol{\Sigma}$ shrinks, shrinks directionally along $\boldsymbol{\phi}$, and beats isotropic $c\mathbf{I}$; order- and re-ask-invariant.	Unusable posterior; scale-only or decorative $\boldsymbol{\Sigma}$.
V4 (R4)	Cold full/tail curve rises to budget on a bank fixed in advance.	Flat model; selection masquerading as ingestion.
V5 (R5)	Sign-flip lowers NDCG; neutralisation costs NDCG; graded beats a fair binarising ablation on the short interview.	Sign-blind and intensity-blind folds.
Integrity items, across all five		
I1	Wrong-user, shuffled, placebo, and duplicated answers gain nothing over the true-answer arm.	Cardinality confound; existential leakage.
I2	Answers come from fold-in behaviour only: no recommender-derived quantity, no target-half information.	Circular answer model; target leakage.

Table 1: **Verification: one check per requirement, plus two integrity items.** Each of V1–V5 is stated so that it can fail, and each certifies exactly one requirement. I1 and I2 apply across all five and exist because the per-requirement checks admit false positives—a sign-blind fold, a decorative covariance, or a circular answer model can pass V1, V4 and V5 while carrying no signal (text). The *ruler* is certified separately by the published-number bridge (§7.2), which is a property of the measuring apparatus rather than of the instrument.

We report *both* full and tail NDCG@10 for every gate (a project reporting rule: full-only or tail-only hides the effect, which is typically much larger on the tail than on the full catalogue). Every per-build gate has been run and passes (Table 8); V1 is reported in full in §7.

3 Related work: a six-family taxonomy

We organise interview-relevant recommenders into six families and score each against R1–R5 (Table 2). The reading is consistent across families: no family is strong on both the accuracy axis (R1) and the interview axis (R2–R5). Figure 1 plots the same fact quantitatively, with measured full-profile accuracy on one axis. Table 2 is the compact conceptual overview; the chapter's anchor is the per-system *master table* (Table 3), which classifies every reasonable candidate against R1–R5 *and*, where one exists, prints its full/tail NDCG@10 on our shared ML-25M ruler—so that the bifurcation is not merely asserted from published claims but is visible in the one column of measured numbers.

No dash is an unexplained omission. Every unmeasured row carries one of three dispositions. (Systems with neither a measurable number nor a distinct capability profile—e.g. SLIM, strictly dominated by its closed-form successor EASE in every published comparison—are cited in the text but given no row.) *Not run here*: GF-CF, BSPM, BERT4Rec and EDDI have public code and are runnable on this ruler, but carry no number in this chapter; the graph-filter family is represented by the measured Turbo-CF row and the sequence family by the measured SASRec row. *Best-effort measured*: for systems whose full-profile core is runnable but whose exact recipe is unpublished or benchmark-bound, we report a best-effort replication of that core with every substitution stated—the node recommender that is the Golbandi tree's core, at 0.3065/0.1895, and the RBMF/Functional-MF seed at 0.2534/0.1764. **One substitution deserves emphasis because it works against the systems it applies to.** The canonical ruler binarises at $r > 3.5$ (§7.2), which is the *published* input for the implicit-feedback family—EASE, EDLAE, the Mult-VAE line, RecVAE, the graph filters and the sequence models were all published on binarised interactions, and feeding them ratings would depart from those systems rather than repair them. Three rows are different: Golbandi's tree routes users by *lovers, haters and unknowns* and is evaluated on RMSE; RBMF/Functional-MF is a rating regression; and TaNP is episodic meta-learning for *rating prediction* over (item, rating) support pairs. All three are ratings-native, and on this ruler all three receive binary input. The mechanism deserves stating exactly, because it is stronger than binarisation: the Liang recipe does not flatten sub-threshold ratings, it *discards* them—every interaction at $r \leq 3.5$ is removed before the matrices are built. A hater is therefore not a zero in these models' input; the hater is absent from it. Golbandi's third branch has nothing to route, the regression fits ones rather than graded targets, and TaNP's rating channel is a constant. This is precisely the information R5 identifies and V5 values at -0.2985 when it is removed from our own tower. **Those three numbers are therefore lower bounds on what the designs can do, and we do not read the gap between them and the tower as evidence about the designs.** TaNP makes the point concretely: at 0.2645/0.1635 it is the strongest of the three on this ruler, reached with its rating channel held constant.

We are equally explicit about the other side of that asymmetry. Our interview does not read the split's fold-in file; it reconstructs each cold user's profile from the *raw* catalogue ratings across all bands, excluding held-out targets, which is how a signed answer becomes available at all (7,563 of the 10,000 test users have at least three sub-threshold ratings). Held-out items are excluded in both cases, so neither input sees its own targets—but the two inputs differ in content and not merely in encoding, and the instrument's is the richer one. Recovering intensity for the ratings-native rows within the existing split is cheap, since the surviving 4.0–5.0 values are in the raw data and the Golbandi node means and RBMF ridge solve consume whatever values the matrix carries. Recovering *sign* is not, because the sub-threshold interactions are not in the split at all: it requires giving those models the same all-bands profile the instrument reads, which changes the input set rather than its encoding. That comparison is the one worth running, and we report it as owed rather than done. The same caution applies to ICF, whose ceiling we take from the measured iALS row: ICF is Bayesian probabilistic MF over ratings, while iALS is implicit-with-confidence, so that substitution crosses an input regime as well as a model family. For Bıyık and ConTS we print a dash rather than a number, because the honest statement is that neither reports full-catalogue accuracy at all—Bıyık's NDCG is agreement between a user's true and estimated top-$|S|$ items over 16 sampled users on slates of 5—and measuring our own reconstruction of a mechanism they share would produce a number that is ours, not theirs. That mechanism is measured in §5, where it belongs: as the cost of ranking on a conjugate posterior mean, which is a decision about *our* architecture. The belief corner is blank in this column because full-catalogue top-N accuracy is not what these papers report—and that is a defect in the record rather than a neutral choice of emphasis, for the reason set out in §4.1. *Not runnable by construction*: PEBOL maintains per-item Beta posteriors over an LLM-scored shortlist of ∼100 items and contains no full-catalogue ranker to run; GATE elicits but does not rank; zero-shot LLM ranking of an 18k catalogue is neither computationally nor methodologically meaningful at full-profile scale (and its collaborative weakness is already documented [17]); BCIE requires its authors' knowledge graph, and a port would substitute so much that the number would no longer be theirs—its conjugate-fold core is covered by the Bıyık-style best-effort arm. ICF is the one row whose full-profile core is *already* measured: its underlying model is Bayesian probabilistic MF, so its full-profile ceiling is the measured iALS row (0.2442).

The measured-accuracy column falls across the belief corner. Read Table 3 down its measured-accuracy column. Every *competitive* number—a full/tail NDCG@10 at or near the frontier on our shared ML-25M ruler—sits in block (a) or (b): the item-only linear and amortised recommenders (RecVAE 0.3540, EASE 0.3476, EDLAE 0.3433). The belief corner, block (c), is either *blank* or *measurably weak*. Both of its runnable cores land well below the frontier—the Golbandi node recommender at 0.3065/0.1895 and the RBMF/Functional-MF seed at 0.2534/0.1764—and both carry an ✗ on R1. The rest of the corner—EDDI, BCIE, UNICORN, PEBOL, GATE, the LLM-zero-shot CRS—has a blank accuracy column entirely. This is not an accident of our harness. Most CRS and elicitation papers never report full-catalogue top-N accuracy at all (they report elicitation RMSE, cold-start MRR/MAP on a shortlist, or critique metrics), and several *cannot produce it by construction*: a per-item Beta posterior (PEBOL) or the decision tree itself (Golbandi, whose node recommender we measure only as a best-effort core) does not induce a ranking over the whole catalogue, and an LLM CRS needs an API call per candidate to score one. Conversely, every system at the frontier is item-only and point-estimate—✗ on R2 and R3. The two blocks are disjoint: no row combines a frontier accuracy number with ✓s across R2–R5. The top-right cell of the gap map—all five requirements met *and* a competitive full-profile number—is empty across the published literature, and is occupied by the instrument this chapter builds. That is the chapter's thesis, and the master table is where it is impossible to miss.

The accuracy corner (families ii–iii). On the canonical ML-20M strong-generalisation split (10k validation / 10k test held-out users, NDCG@100), the full-profile frontier has been *flat since 2019–2020* and is held by shallow models, not deep nets: EASE [40] at NDCG@100 0.420, Mult-VAE [25] at 0.426, RecVAE [38] at 0.442 (top of this split), with EDLAE [41] in the same class; SLIM (0.401) and WMF/iALS (0.386) are the classic floors. Dacrema et al. [12] is the binding warning: most deeper "SOTA" gains vanish on this exact split. The graph-filter leaders (GF-CF [37], BSPM [8], Turbo-CF [32]) are the 2023–24 speed+accuracy front on Gowalla/Yelp/Amazon but do not report this ML-20M split, so they are admissible to our accuracy tier only if re-run on our harness. Turbo-CF is, and lands at 0.2622/0.1621—mid-pack rather than frontier at this catalogue size. All of family (ii)/(iii) are item-only: their input is a one-hot (or set-indicator) interaction vector, which *cannot encode* a signed value, an out-of-catalogue concept, or a continuous direction—R2 and R5 are an architecture property here, not a tuning gap.

The belief corner (family iv, and vi). Bıyık et al. [5] maintain a multivariate-Gaussian belief over a user embedding and treat both item comparisons and "soft attributes" (e.g. "scarier") as directions (concept-activation vectors) in the item space, selecting queries by expected value of information—the nearest thing to R2+R3+R4 together. ConTS [23] unifies items and attributes as Thompson-sampling arms with a posterior over a cold-start user. BCIE [44] folds critiques into a *frozen* factorisation by exact conjugate-Gaussian updates—the closest published analogue of our belief mechanism. PEBOL [3] keeps a per-item Beta posterior updated by LLM-extracted aspects and NLI scoring, reporting cold-start MRR@10 0.27 versus a 0.17 baseline. None of these reports full-profile accuracy at the EASE/RecVAE bar; each is a belief layer, not a certified tower.

The set-encoder and cold-start-NP precedents (families i, iii). EDDI/Partial-VAE [30] is the permutation-invariant set-encoder over an arbitrary observed subset with information-gain acquisition—exactly R2's architecture, but demonstrated outside recommendation and without a full-profile CF result or an attribute/concept channel. TaNP [26] gives a neural-process, stochastic-process uncertainty over a value-carrying set input for cold-start—the R2+R4 shape, but not at full-profile SOTA nor channel-agnostic. These are cited as the mechanism precedents our design instantiates, not as rivals for the conjunction.

The gap map. Figure 1 places every system on two axes: *interview-compatibility* on the x-axis—a coarse ordinal capability count, the number of **R2–R5** marks a system satisfies in the master table (R1 is excluded because accuracy is the y-axis)—and *measured full-profile NDCG@10 on our ML-25M ruler* (R1) on the y-axis. The population is bimodal: the accuracy corner collapses onto $x \leq 1.5$, the interview-capable systems sit at $x \geq 2$ with weak or non-existent full-profile accuracy, and no system bridges—the top-right, high compatibility *and* frontier accuracy, is empty.

4 Gap analysis

The verdict. No single published system satisfies all of {R1 SOTA-full-profile $\wedge$ R2 any-length set $\wedge$ R3 uncertainty-native $\wedge$ R4 monotone $\wedge$ R5 expressive answers}. The binding near-miss is Bıyık et al. [5]: it fails R1 (a d=50 ALS two-tower rather than a certified RecVAE-class tower, and it reports no full-catalogue accuracy—its NDCG is computed between a user's true top-$|S|$ items and the top-$|S|$ estimated from the posterior, over 16 sampled test users on slates of 5, an in-simulator alignment measure) and gives only an approximate, "generally not Gaussian" posterior; it fails R2 in the specific sense of §4 (a per-concept supervised probe over a closed tag list, not an open vocabulary). Every other candidate fails R3 or R4 outright, and we are aware of no system that occupies the conjunction.

Where each near-miss falls short.

- **Bıyık et al. 2023** [5] — holds *part* of R2 (items and soft attributes both enter as directions) and R3 (a Gaussian belief), over a *frozen* item tower—they train the CF model and the attribute directions separately, as we do. *Fails R1*: a d=50 ALS two-tower, and accuracy is reported on their own elicitation task (in-simulator top-$|S|$ agreement over 16 users, slates of 5), not the EASE/RecVAE bar. *Fails R2* in the sense that decides it: the directions are *per-attribute supervised probes* over an enumerated tag list—each a logistic regressor fitted from labelled positive/negative items, trained for the 164 most-frequent tags—so the vocabulary is fixed before the user arrives, which is exactly what R2 forbids. R4 is approximate (sampled EVOI over a non-Gaussian posterior). This is the central threat, and our wedge is exactly (a) an *open* vocabulary through one global adapter with no per-concept supervision, (b) the R1 accuracy bar on a shared full-catalogue ruler, (c) exact conjugacy vs. approximate posterior.

- **ConTS** [23] — R3 + a unified item/attribute action space. *Fails R2* (a fixed categorical attribute schema; no arbitrary embedding, no continuous or open token) and *R1* (a bandit over arms, not a full-profile encoder).
- **EDDI / Partial-VAE** [30] — R3 (information-gain over a belief) + set input. *Fails R1* (not a recommendation model / no CF SOTA) and *R2* (a fixed feature set; no attribute or concept channel as demonstrated).
- **RecVAE / EASE** [38, 40] — *R1 yes. Fail R2* (an item-only one-hot / indicator basis admits no concept and no continuous direction), *R5* (nor a signed or graded value), and *R3* (point estimate; RecVAE's latent posterior is item-only and is not used as an actionable belief).
- **PEBOL** [3] — R3 (per-item Beta) + R2 (LLM-extracted aspects, an open vocabulary) + strong cold-start. *Fails R1* (no full-profile CF ranking) and its belief is *per item*, not a shared user covariance; no monotonicity guarantee, and a yes/no answer carries no intensity (R5).
- **SASRec / BERT4Rec** [20, 42] — re-encode the sequence at inference. *Fail R2, R3, R4 and R5* (item-only, order-sensitive, point estimate, non-monotone—a later token can reorder everything). Token input is trivially satisfied but is *not* value-carrying or mixed-type, so "interview = tokens" does not hold; and BERT4Rec's fold-in advantage carries a replicability caveat [34].

The three inexpressible legs. Three input channels are *architecturally impossible* (not merely untuned) in the item-indicator basis of every R1-bar model: (i) graded/signed per-entity values (the basis has no sign and no magnitude—a "dislike" is not representable as a negative one-hot without changing the likelihood), (ii) concepts whose vocabulary is not fixed when the model is fitted (a column exists only for an entity that was present at fit time), and (iii) arbitrary continuous direction tokens. Folding these as *linear observations in a frozen latent* is the one leg of the conjunction we find nowhere in the literature.

The distinction that matters is between an *open* and a *closed* channel. An item-indicator model can be extended with a concept—as a pseudo-item column, or an extra input unit—but only by adding a column, and a column is fixed when the model is fitted (§7). Its concept vocabulary is therefore frozen at training time. Our interface takes a direction $\boldsymbol{\phi} \in \mathbb{R}^d$ and a value, so any entity with an embedding is a legal observation at inference: a paraphrase, a concept coined after deployment, or a continuous direction emitted by a policy and snapped to a nearby meaningful axis. This is what we mean by channel-agnostic, and it is a property of the interface rather than a quantity to be tuned.

4.1 Feasibility and necessity of the conjunction

One objection to the analysis above is that the conjunction is inexpensive to satisfy: take a frontier recommender, carry a conjugate belief beside it, and the grid fills in. Three considerations tell against this.

Most of the frontier has nothing to carry a belief over. A belief of the form Eq. (2) needs a user vector in a dense latent, and a direction in that latent to be a meaningful query. EASE has neither: its parameters are an item–item weight matrix and its "user" is the raw interaction row, one coordinate per catalogue item. There is no low-dimensional taste vector to be uncertain about and no direction to read $w^\top \boldsymbol{\Sigma} w$ along. The same holds for EDLAE and for the training-free graph filters. For the closed-form family—most of the accuracy corner—the construction is not difficult, it is undefined. It is available to the amortised-encoder family, and that is where we build it.

Where it is available, it carries a cost. The layer has to leave full-profile accuracy exactly intact, which is a property to be measured and not assumed: our empty interview reproduces the frozen decoder's ranking identically (Spearman 1.0000, $|\boldsymbol{\mu}_0| = 0$), and the obvious alternative—ranking on the conjugate mean—forfeits 0.15 full NDCG (§5). Nor does a dense latent accept concepts merely because it is dense: a raw concept centroid ranks its own member items at AUC 0.44, *below chance*, and only whitening makes it a direction (0.94). The channel is built, not inherited.

And the reason to pay is that policy conclusions are not portable across recommenders. This is the argument for an instrument, and it is empirical rather than rhetorical. On a strong ingestion model, differences between question-selection strategies compress: a *random* question from an answerable bank matches popularity-ranked items at $q{=}1, 2, 4, 8$ and loses only at $q{=}16$; a one-level adaptive probe pays +0.0015 full (CI $[+0.0007, +0.0024]$); and $\boldsymbol{\Sigma}$-greedy beats random by +0.0036 at the largest budget alone. A weaker recommender leaves more of the work to the question, so the same strategy gap measured there is larger—and, if strategies are affected unequally, need not even preserve the *ordering*. A policy result obtained on a weak recommender therefore carries an unknown amount of its own substrate, which is exactly the confound this chapter exists to remove.

On what the elicitation literature reports. We are direct about this because it bears on the table above. Reporting cold-start MRR over a hundred-item shortlist, or top-$|S|$ agreement between a simulated user and a posterior, or turn counts to a success state, measures a system on a task that is not the one it would perform in service. Full-catalogue NDCG is not a formality: it is what a user experiences when a deployed system ranks the whole catalogue for them after an interview, and it is the only number in which an elicitation gain and a recommender weakness can be compared on the same scale. A shortlist metric can look excellent while the underlying model could not rank the catalogue at all, and a policy improvement measured on it may not survive contact with 18,359 candidates. We therefore treat the blank accuracy column not as a difference of emphasis but as missing evidence, and we report full *and* tail NDCG@10 for every arm of our own work, including the arms where it is unflattering.

Prior art, cited and not claimed. We must credit, and do not claim priority over: attribute-answers-as-directions plus a Bayesian fold [5]; unified item+attribute belief and value-of-information [5, 23]; conjugate-Gaussian folds over a frozen factorisation [44]; permutation-invariant set-encoders with information-gain acquisition [30]; value-carrying set-input cold-start recommenders [26]; the non-degradation-of-information principle [15, 6]; and the now-crowded "uncertainty-native recommender" area [7]. Our defensible novelty is only the *conjunction* of all five requirements in one frozen certified tower, plus the three inexpressible legs and the acceptance battery that certifies them.

5 The instrument

The design follows directly from the gap analysis: use the shallow frozen tower that already holds the accuracy corner (R1, and the flat-since-2019 frontier means no deep-net detour is warranted), and add the one belief mechanism with a principled monotonicity story—exact linear-Gaussian conjugacy—over its frozen latent (R3, R4), with a channel-agnostic linear-observation interface (R2). The tower half is built and certified (§7); the belief layer is derived here, fitted on real data, and gated in §7.3.

5.1 Frozen shallow-SOTA tower

The substrate is a certified shallow recommender—a RecVAE-class amortised encoder or a closed-form autoencoder [38, 40]—trained once on the full catalogue and then *frozen*. It exposes (a) item factor rows $\mathbf{Q}_i \in \mathbb{R}^d$, (b) a decoder read-out $s_i(\mathbf{u}) = \mathbf{Q}_i^\top \mathbf{u} + \beta_i$ with a learned bias β_i, and (c) the encoder that maps a full interaction set to a latent. Freezing is the guarantee that V1 (full strength preserved) is a property of the substrate, not of the belief layer. It is a design choice we share with the soft-attribute line rather than one that distinguishes us: Göpfert et al. are explicit that "we do not use the tag data when training the collaborative filtering model" [16], and Bıyık et al. train the CF model and the attribute directions separately [5]. We cite that convergence as independent support for the frozen-substrate design. We deliberately do *not* extend the linear basis with concept columns (that stays in the item-indicator world and fails R3), nor learn the uncertainty head (no monotonicity story, and retraining risks V1). The tower is now *selected and certified*: the graded-native set-encoder T2′, trained once in its pre-registered configuration, scores 0.3482/0.2462 full/tail on the held-out test cohort—a paired-bootstrap tie with the 0.3540/0.2497 RecVAE bar—and reproduces the frozen decoder exactly at the empty interview (§7). It is this checkpoint that the belief layer below is placed over.

5.2 Analytic conjugate-Gaussian belief layer

Over the frozen latent we place a Gaussian belief on the user vector, $\mathbf{u} \sim \mathcal{N}(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0)$, and model each answer as a noisy linear read-out along the queried direction:

$$y_j \;=\; \boldsymbol{\phi}_j^\top \mathbf{u} + \varepsilon_j, \qquad \varepsilon_j \sim \mathcal{N}(0, \sigma_j^2). \tag{1}$$

Because the observation is linear-Gaussian, the posterior after $\mathcal{E}_t$ is Gaussian in closed form; maintaining it in *precision* form (a precision accumulator) makes the any-length fold-in a single running sum:

$$\boldsymbol{\Lambda}_t \;=\; \boldsymbol{\Lambda}_0 + \sum_{j=1}^{t} \sigma_j^{-2}\, \boldsymbol{\phi}_j \boldsymbol{\phi}_j^\top, \qquad \boldsymbol{\Lambda}_t\, \boldsymbol{\mu}_t \;=\; \boldsymbol{\Lambda}_0 \boldsymbol{\mu}_0 + \sum_{j=1}^{t} \sigma_j^{-2}\, y_j\, \boldsymbol{\phi}_j, \qquad \boldsymbol{\Sigma}_t = \boldsymbol{\Lambda}_t^{-1}. \tag{2}$$

Equation (2) is exactly the mechanism BCIE [44] uses for critiques over a frozen factorisation; our contribution is doing it over a *certified full-profile SOTA* tower with channel-agnostic tokens, and certifying it against V1–V5. Three properties fall out by construction: it is a permutation-invariant *set* operation defined for every $t \geq 0$ (R2); each update adds a rank-1 term along $\boldsymbol{\phi}_{t+1}$, so $\boldsymbol{\Lambda}_t$ only grows and $\det \boldsymbol{\Sigma}_t$ only shrinks (V3); and conjugate updates are monotone in *expected utility* [15, 6]. The last is why R4 is plausible, but—because the tower's dot-product ranker does not act optimally under the belief—per-question NDCG monotonicity is *not* automatic and must be shown empirically (§8).

A single answer tightens a correlated region, which accounts for coarse-to-fine ordering. Because each update adds a rank-one term, the Sherman–Morrison identity gives the effect of asking $\boldsymbol{\phi}$ on the uncertainty along *any other* direction w in closed form:

$$w^\top \boldsymbol{\Sigma}_t w \;-\; w^\top \boldsymbol{\Sigma}_{t+1} w \;=\; \frac{\sigma^{-2} \left(w^\top \boldsymbol{\Sigma}_t \boldsymbol{\phi}\right)^2}{1 + \sigma^{-2}\, \boldsymbol{\phi}^\top \boldsymbol{\Sigma}_t \boldsymbol{\phi}}. \tag{3}$$

The reduction is proportional to $(w^\top \boldsymbol{\Sigma}_t \boldsymbol{\phi})^2$—the squared covariance between w and the queried direction under the current belief. A question therefore does not sharpen one axis; it sharpens a whole *correlated region*, in proportion to how strongly that region moves with what was asked. The consequence for a mixed question bank is immediate and geometric rather than heuristic: a concept direction is by construction close to many item directions at once, so folding it collapses a large region per question, whereas an idiosyncratic item collapses little beyond itself. Breadth-before-depth is thus a property of the covariance geometry, not a scheduling rule imposed from outside—and it is what we observe when a purely accuracy-driven greedy is allowed to choose freely from items and concepts together (§6.4). Coarse-to-fine itself is long known in the interview literature [14, 36]; what Eq. (3) adds is an account of *why* it holds in a latent-factor instrument, and a way to compute it for any channel.

The belief separates cleanly into an answer-independent and an answer-dependent part. A property of Eq. (2) deserves to be stated explicitly, because anyone building elicitation on this instrument needs it. The precision $\mathbf{\Lambda}_t$ accumulates $\sigma_j^{-2}\boldsymbol{\phi}_j\boldsymbol{\phi}_j^\top$ terms: it depends on *which* directions were queried and on the noise assigned to each, but *not on the answers themselves.* The covariance $\mathbf{\Sigma}_t$ is therefore the same for two users who were asked the same questions and replied differently. The mean is not: $\boldsymbol{\mu}_t$ carries $\sum_j \sigma_j^{-2} y_j \boldsymbol{\phi}_j$ and moves with every y_j.

This asymmetry is informative for elicitation. A question-selection rule that reads only the covariance is, by construction, answer-independent—it produces the same sequence for every user, which is the linear-Gaussian, variance-only case in which adaptive selection provably gains nothing over a fixed schedule [21, 19]. A rule that reads the mean, or any quantity derived from it such as the current ranking, is not. Our instrument exposes both, so the choice is available to a policy designer rather than forced by the substrate; which rule to use, and what it is worth, is the companion chapter's question.

One qualification, which the refusal band supplies. The per-observation noise σ_j is not constant across answer types: an unanswerable question is not folded at all, so whether a user *could* answer changes which terms enter $\mathbf{\Lambda}_t$. The covariance is consequently answer-dependent to the extent that answerability varies across users—and it varies a great deal, since items and concepts are answered at very different rates (§6.4). Answerability thus enters the belief itself, not only the accounting of a wasted turn.

What the belief layer provides. The layer is built and fitted: $\mathbf{\Lambda}_0, \boldsymbol{\mu}_0$ are taken from the tower, and the per-channel noise reduces to eight fitted scalars (a 2×3 channel×confidence precision grid plus a prior scale and a variance floor), calibrated on real data with a likelihood-ratio sign proof on held-out concept triples. It passes V3, V4 and V5 (Table 8): the posterior shrinks monotonically over answers, shrinks 11.5× more along the queried direction than elsewhere, is order-invariant to 5×10^{-7}, and is unmoved by a refusal. Uncertainty along any direction w—an item, a concept, a phrase—is read off as $w^\top \mathbf{\Sigma}_t w$.

That is the instrument's obligation discharged: a belief that is correct, queryable for any channel, and available to a policy. What a policy should do with it—and in particular whether reading the covariance is a good way to choose questions—is a separate question, and the companion chapter's subject.

5.3 Channel-agnostic tokens: one representation, one form of observation

Every channel supplies the same kind of thing—a direction $\boldsymbol{\phi}_j \in \mathbb{R}^d$ in the frozen latent and a value $y_j \in \mathbb{R}$—and it is this that makes the channel set open (R2). Two folds consume those pairs in the system we report: the tower's own set encoder for item evidence, and a trained gated operator for concept evidence (§6.2); the conjugate update of Eq. (2) maintains the belief over the same latent alongside them. We are explicit that these are distinct operators rather than one code path, because the unification that matters is of the representation: any entity with an embedding is expressible as an observation, which is what no item-indicator model can say.

Why the belief is maintained alongside the ranking rather than inside it. The conjugate posterior mean is itself a legal ranker—score item i by $\mathbf{Q}_i^\top \boldsymbol{\mu}_t + \beta_i$—and it would make Eq. (2) the single fold for every channel. We measured that option rather than assuming it: a conjugate-Gaussian belief over a frozen factorisation, ranked by its posterior mean, reaches full/tail NDCG@10 0.2019/0.1200 on our ruler, against 0.3482/0.2462 for the learned fold (Table 7). Ranking on the posterior mean therefore costs roughly 0.15 full NDCG—it forfeits R1 to buy uniformity—and that is why the belief is carried beside the ranking path and read by a policy, not used to produce the rankings we report.

The channels, then, are:

- **Items.** $\boldsymbol{\phi}_j = \mathbf{Q}_i$ (the frozen item factor). A graded or signed answer sets y_j; a dislike is a negative y_j along the same direction—representable here precisely because the observation model (1) has a sign and a magnitude, unlike a one-hot indicator.
- **Out-of-catalogue concepts.** $\boldsymbol{\phi}_j$ is a concept direction in the latent (e.g. a centroid of member factors, or a text-adapter image of the concept), folded exactly like an item—no catalogue column required.
- **Entities / free text.** An embedding is projected into the latent by a fixed adapter and folded identically.
- **Continuous direction tokens.** A policy may emit any $\boldsymbol{\phi}_j \in \mathbb{R}^d$ directly (the continuous-action interface the companion chapter uses); the belief update does not distinguish it from a catalogue entity.

The signed/graded, out-of-catalogue, and continuous legs are the three channels inexpressible in the R1-bar basis (§4).

Any direction is a legal observation: an illustration. The signed/graded leg is delivered and measured (V5, Figure 2). The concept leg is delivered over the genome vocabulary. The third leg—an *arbitrary* direction, including one for an entity that has no catalogue column and was never seen in training—is a property of the interface rather than of any particular channel: everything downstream of Eq. (2) consumes a pair $(\boldsymbol{\phi}, y)$, so any entity with an embedding is admissible. We illustrate that concretely.

Each of the 1,031 genome concepts has a direction in the latent and a name in English, so those pairs determine a single global linear map from a frozen general-purpose sentence encoder into the frozen latent, in closed form. We fit it on a random 80% of the concepts and ask what happens to the 20% it never saw: for a held-out concept, the direction predicted *from its name alone* recovers the top of that concept's own item ranking more than 300× more often than chance (0.167 top-10 overlap against a 0.0005 random baseline; cosine 0.638 to the true direction). No per-concept probe is fitted, no inventory is consulted, and nothing is retrained—a phrase becomes a direction by one matrix multiplication.

Because the map is defined on any string, it also accepts phrases that are not in our concept vocabulary. To make that meaningful rather than decorative, each phrase below is screened against the vocabulary two ways—no genome tag occurs as a whole-word substring of it, and no concept sits within cosine 0.6 of it—and we print the *closest* concept we actually hold, so the reader can

see how far the phrase sits from anything curated. (The screen is not cosmetic: it rejects "revenge served slowly", "loneliness in a big city" and "a detective who drinks too much", because *revenge*, *loneliness* and *detective* are all themselves genome tags.)
The systematic claim is the held-out number above; the table shows what it buys. The last row makes the point plainest: the nearest concept in the entire vocabulary to "movies about grief" is *feel good movie*, tonally its opposite, and the phrase's direction shares none of its top ten items with that concept's curated direction—the phrase is placed by its own meaning, not snapped to the closest thing on a list. The phrases were drawn from a screened pool, and ones whose meaning the genome vocabulary does not carry at all—"set entirely inside one room"—return lists we would not defend.

We present this as an *illustration of what the interface permits*, not as a contribution to mapping language into collaborative latents—a question with an established literature of its own, in which attribute directions are obtained either by per-concept supervised probes [16, 5] or by per-attribute corpus retrieval [4]. Fitting such a map *well* is a research direction this architecture opens rather than one this chapter pursues: text underdetermines collaborative taste, and how much of that gap a better-fitted adapter can close is exactly the kind of question the instrument now makes measurable.

The observation is independent of its provenance. One consequence is worth naming because it requires no additional mechanism. The discussion above treats a phrase as something the *system* elected to ask about. But Eq. (1) reads a pair $(\boldsymbol{\phi}, y)$ and is indifferent to its provenance: a user who volunteers "I'm in the mood for something about grief" has supplied a direction and a positive value, which folds exactly as an answer to a question we posed would. The same holds for a critique after a recommendation ("less bleak than that") and for a direction a continuous policy emits and never renders as language at all. An item-indicator model cannot accept any of these, because none has a column; a model with a curated attribute list can accept only those a curator anticipated. A user-initiated request is thus not a separate feature to be built on top of this instrument—it is the same fold read in the other direction, and the reason it is available is R2. We have not evaluated proactive requests here, and the quality of any such interaction would rest on the text adapter whose limits are set out above.

6 The concept channel: derivation, lineage, and deployment currency

The out-of-catalogue concept leg is the most demanding of the three inexpressible channels—it is the one with no catalogue column, no interaction count, and no sign in the item-indicator basis—so we work it out in full. Three things must be made concrete: how a concept answer becomes an observation pair $(\boldsymbol{\phi}_c, y_c)$ for the belief update of Eq. (2); where each piece sits in the published record (so that we claim only the conjunction, never a component); and why this channel is the one that forces the battery to gate what a *deployed* interview elicits and not only what the operator can express. This section is written at derivation level; the concept-channel numbers below are measured on the internal ML-25M interview harness and are reported as such.

6.1 A signed, exposure-corrected affinity estimator

A concept answer must supply the value y_c that Eq. (1) reads along the concept direction. We want that value to be *signed* (a user can dislike "horror," not merely fail to like it), *popularity-*

corrected (a concept made of blockbusters must not read as universal affinity), *volume-corrected* (a heavy watcher must not read as liking everything), and *confidence-bearing* (a concept the user has plausibly never been exposed to is a *refusal*, not a dislike). Two estimators, an implicit and an explicit one, combine to meet this.

Let $\mathcal{I}_u$ be the user's watched set, p_i the catalogue marginal of item i, and let concept c have member set M_c. Write $n_c = |\mathcal{I}_u \cap M_c|$ for the user's watched members and $e_c = |\mathcal{I}_u| \cdot \pi_c$ for the number *expected* under the user's volume times the concept's popularity $\pi_c = \sum_{i \in M_c} p_i$. The implicit *selection* estimator is the add-$\frac{1}{2}$ log watch-lift

$$\mathrm{SEL}(u, c) \;=\; \log_2 \frac{n_c + 0.5}{e_c + 0.5} \;\;\approx\;\; \mathrm{PMI}(u, c), \tag{4}$$

which is precisely the *pointwise mutual information* between the user and the concept: the denominator's π_c divides out concept popularity, and forming e_c from the user's own volume divides out user activity, so SEL is positive for genuine affinity, near zero for popularity-explained co-occurrence, and *negative for avoidance.* Its normalised form is the NPMI, $\mathrm{NPMI} = \mathrm{PMI}/(-\log_2 \hat{p}_{uc}) \in [-1, 1]$, which tames PMI's small-count inflation and yields a bounded, comparable score. The explicit *value* estimator is a support-shrunk signed rating residual,

$$\mathrm{VAL}(u, c) \;=\; \frac{n_c}{n_c + \lambda}\, \bar{r}^{\,\mathrm{res}}_{uc}, \qquad \lambda = 3, \qquad \bar{r}^{\,\mathrm{res}}_{uc} = \frac{1}{n_c} \sum_{i \in \mathcal{I}_u \cap M_c} \big(r_{ui} - \bar{r}_i\big), \tag{5}$$

the mean rating the user gives the concept's members *above each item's own mean*, shrunk toward zero by the Beta-style factor $n_c/(n_c+\lambda)$ so that a one-film opinion counts for little. Both estimators are signed and combine into a four-band answer: a *graded like* (SEL/VAL band above the neutral zone), a *neutral* (indistinguishable-from-exposure), a *graded dislike* (band below), and a *refusal* issued exactly when the expected support falls below a threshold, $e_c < \tau$, i.e. when the user has plausibly never been exposed to the concept and no answer should be forced.

Lineage. Every piece is established prior art. The log-lift (4) is PMI/NPMI, the standard popularity-corrected association score. Reading a *magnitude* of implicit consumption as evidence is Hu–Koren–Volinsky's implicit-confidence template [18]—except that HKV lets the count re-enter *only as a loss confidence weight, never as a sign*, which is exactly the assumption our signed estimator breaks. The principled account of why a non-watch is not a dislike is ExpoMF [24]: it separates a latent *exposure* variable (was the user even aware of the item) from *preference*, so that non-consumption is modelled as "not exposed" *or* "exposed-and-disliked"—the generative gold standard our refusal band ($e_c < \tau$) and popularity-corrected e_c approximate cheaply and in closed form. The unbiased-learning route to the same correction is attribute-level exposure propensity [35], which IPS-reweights implicit feedback by a per-attribute propensity; it is the heavier, variance-prone alternative to our moment estimator. Finally, calibrated recommendation [39] builds a user's normalised genre *distribution* $p(g \mid u)$—the closest widely-used "concept profile," but a distribution that sums to one and therefore encodes relative emphasis only: it *cannot express dislike*, which is precisely the expressive gap SEL< 0 fills.

Why the implicit channel, quantified. On an informativeness decomposition (a per-concept linear fit of the oracle concept affinity onto each candidate answer, R^2), the implicit watch-lift dominates: SEL explains 0.376 of the affinity variance, the explicit rating residual VAL a further +0.042 (SEL+VAL 0.398), while an LLM-elicited ordinal answer explains only 0.017 (an internal decomposition)—implicit outperforms the explicit LLM answer by roughly 22×. Because SEL is built from raw watch counts and item marginals, it uses no recommender geometry, so this signal is *non-circular by construction* (it even beats a model-geometry projection of the same residuals). This is the evidence that the concept channel's value is worth carrying signed rather than as a mere membership indicator; the decomposition is an informativeness measurement, not an interview curve, and is reported as such.

6.2 The fold operator: whitened directions and a gated fold-to-point

The direction $\boldsymbol{\phi}_c$ must point where the concept's taste lives, not where popularity lives. A raw member centroid fails this: a shared popularity component (random concept pairs have cosine ≈ 0.5) masks the genre geometry. Removing it—centre the member factors and strip the top-1 principal component (the popularity axis), the all-but-the-top whitening of Mu and Viswanath [31]—turns the centroid into a usable direction: concept-membership discrimination through the frozen decoder rises from AUC 0.44 (raw, at chance) to 0.94 (whitened). We therefore take $\boldsymbol{\phi}_c$ to be the *whitened member centroid*, IDF-weighted by $1/\log(1+|M_c|)$ so broad genres do not dominate. Whitening is a fix, not a contribution, and is cited as such.

The operator that consumes $(\boldsymbol{\phi}_c, y_c)$ is where the design decision lives, and the two candidate families separate cleanly under ablation. An *additive-union* operator scores each item by summing concept contributions on a popularity floor, $s_i = \beta_i + \sum_c w_c \langle \mathbf{W}_d\mathbf{Q}_i, \boldsymbol{\phi}_c\rangle$. It has the right monotonicity floor (an empty interview scores exactly β_i) but the wrong composition: it scores the *union* of directions, so correlated sibling concepts ("dark," "gritty," "noir") re-add the same acclaim axis and over-count. Measured against a no-answer intercept of 0.1279 full NDCG@10, the additive channel peaks early (about +0.011 at two answers) and then *craters* as answers accumulate, reaching 0.0991 at eight answers—below the intercept. A *trained gated fold-to-point* operator fixes this by folding the answers into the latent itself and letting a frozen decoder apply the full collaborative (intersection) structure:

$$\mathbf{u} \;\leftarrow\; \mathbf{u} + \sigma\big(g([\mathbf{u}; \boldsymbol{\phi}_c; y_c])\big) \;\odot\; h([\mathbf{u}; \boldsymbol{\phi}_c; y_c]), \tag{6}$$

with the gate g and the delta MLP h *zero-initialised* (so at initialisation $\mathbf{u}$ is unchanged and V1 survives) and the accumulated shift norm-capped per user. In the concept-operator ablation it is monotone in the number of answers to eight (0.1759/0.0841 full/tail at eight answers, up from 0.1322/0.0503 at one) and, crucially, *redundancy-robust*: feeding it eight *correlated* top-SEL concepts still reaches 0.1685/0.0906, above its four-answer level (0.1572/0.0777), exactly where the additive-union operator degraded. The gated fold is the concept operator we adopt; Eq. (6) is the concept-channel instantiation of the linear-observation fold, with the belief covariance maintained analytically as in §5.

Lineage. The skeleton—an attribute as a latent direction, folded by a Bayesian belief update over a frozen factor space—is well-trodden, and we claim none of it in isolation. Bıyık et al. [5] treat a soft attribute as a concept-activation-vector direction and fold signed answers into a Gaussian belief; M&Ms-VAE folds a rating-posterior and an attribute-posterior over one latent by a product-of-experts merge [1]; its successor adds an independent *signed* critique axis through learned positive/negative conditioning tokens, and reports that negative critiquing saturates after about five turns [2]—a concrete ceiling for any signed-attribute channel, consistent with our own front-loaded concept curves. BK-VAE folds keyphrase-activation-vector directions by a Bayesian critiquing step [29]; the gate/MLP of Eq. (6) is a per-token generalisation of FiLM conditioning [33], which a single global affine cannot do because it would apply the same modulation to item and concept tokens alike. And the decision to *freeze* the tower rather than co-train a concept head is a direct reading of the cautionary result that a bolt-on critique head costs recommendation accuracy [28]. What none of these holds jointly—and what we are aware of no system holding—is the conjunction this channel occupies: *continuous, out-of-catalogue concept directions* (a whitened centroid, or a text-adapter image, with no per-attribute probe to train), driven by a *signed, exposure-corrected implicit affinity* (Eqs. (4)–(5)), folded over a *frozen, certified full-profile SOTA tower*. The components are borrowed; only their conjunction is claimed.

6.3 Deployment currency: why a capability gate is not enough

A gate on what an operator *can* express must be paired with a gate on what the deployed system *does* express, and the concept channel is where the distinction bites. The fold of Eq. (6) is sign-capable: negating every *concept* answer moves NDCG by -0.2012 with a clean CI, so the concept channel passes V5's sign-flip test on its own operator. But that certifies the operator, not the interview. Driven down a realistic, user-independent question bank, most concept answers are weak positives, and an answer model that never emits a negative can accumulate them until the ranking drifts *below* the no-answer intercept while every capability gate still reads green.

We therefore evaluate the channel in *deployment currency*: a fixed, user-independent bank under a per-question budget where every question spends budget whether or not it is answered, scored at the endpoint against the intercept (V4's fixed-bank setting). The acceptance form of the concept channel is not "can the operator express dislike" but "does the realizable interview stay above the no-answer intercept at budget." This is the criterion the signed four-band estimator of §6.1 is built to satisfy, and §6.4 reports it.

6.4 Results: the signed retrain, and where the concept channel wins

The signed retrain of §6.3 has since landed, and we ran the two channels—concepts and items—through *one* deployment-currency harness (10,000 held-out cold users, per-question budget, every question spending budget whether answered or not, scored on full *and* tail NDCG@10 against the no-answer intercept 0.1279/0.0192). They tell a consistent story: the signed answer keeps the deployed interview above the intercept, and the concept channel's value is *shaped*—it owns the opening of an interview and cedes the long tail of it to items—and its largest prize is one we do not yet capture.

The signed answer turns the deployment decline into a gain. Re-running the pre-registered triple gate on the *retrained* model (not merely the eval-only fold of §6.3) clears all three taste criteria. Signed answers beat the positive-only clip by $+0.0104$ full NDCG@10 at the sixteenth question (CI $[+0.0075, +0.0134]$): the clip-up curve sinks below the intercept by the eighth question and stays there (0.1268 at $q{=}16$), while the signed curve holds above it (0.1372 in this gate's arm). The counterfeit control—an answer built only from user volume and concept popularity—reproduces *none* of the gain (it declines, -0.045 at $q{=}16$), so the signal is taste, not a volume$\times$popularity artifact; and permuting the answer values across concepts collapses the gain ($+0.0335$ CI-excluding-zero relative to the permutation), so it is the answer *values* that carry it, not fold cardinality. The two controls together establish what the channel is doing: the gain is carried by the answer *values*, and it is not a restatement of how much the user has watched.

Both channels carry an interview. The instrument's claim on the elicitation side is deliberately modest, and it is worth separating from the claim we do *not* make. What the instrument chapter needs is that *both* channels fold and both move the ranking off the cold intercept—which is what makes the instrument usable for measuring policies over a mixed question bank. That much is unambiguous: an interview conducted entirely in out-of-catalogue concepts, a channel no R1-bar baseline can express at all, lifts a frontier-accuracy recommender substantially and monotonically above its 0.1279/0.0192 starting point, and so does an interview conducted entirely in items.

What we deliberately do *not* state here is a ranking between the two channels, and the reason is methodological rather than diplomatic. Any such comparison is only as meaningful as the question-selection strategy each channel is given, and pairing one hand-picked heuristic per channel—a popularity-ordered item bank against a polarization-ordered concept bank, say—measures the two heuristics at least as much as it measures the two channels. We therefore compare the channels under a single strategy-free criterion: the *best static question sequence* for NDCG over an items-only bank, a concepts-only bank, and a combined bank of both, with the sequences constructed on the validation cohort and scored on the disjoint test cohort. Every bank meets the same criterion on the same cohorts at the same budgets, and the combined bank *contains* the items-only bank, so what separates the arms is the channel rather than whichever heuristics happened to be paired. All three arms are in Table 5. The answerability figure underneath them is the striking one: across a sixteen-question interview a user answers on average only 3.6 of the item questions, the rest being items they have not seen. That is the structural pressure the concept channel exists to relieve. Three things follow.

Concepts own the opening, on both metrics. At one question the concept bank leads on full (.1436 vs .1405) and on tail (.0323 vs .0275), and the combined bank is *identical* to concepts-only through q=4—given free choice over five hundred items and five hundred concepts, the optimiser spends its first four questions on concepts. The emergent combined order is `c c c c i i i i c i i i i i i i`: concepts first, then items, with one concept returning at position nine. Nothing in the objective encodes “ask broad things first.” Coarse-to-fine is long known in the interview literature [14, 36] and we claim no discovery; what is new here is that it emerges unprompted in a *mixed* action space, and that §5 gives a geometric account of why (Eq. (3)).

Concepts dominate the tail at every budget, without being asked to. At q=8 the concept bank reaches .0698 tail against items’ .0538, a 30% margin, and the combined bank is the best tail arm at q=16 (.0776 vs .0679, +14%). **The greedy maximised full NDCG@10 only**—the tail was never an objective in any arm—so the tail advantage is a property of the channel, not something the optimiser was steered toward.

Items win full accuracy on the long interview. From q=8 the item bank leads on full (.1863 vs .1776), and at q=16 combined trails items-only by -0.0049. Neither figure is an artifact of the construction cohort: build and held-out curves differ by -0.0047, -0.0002 and -0.0025 across the three arms, so the test cohort scores *higher* than the cohort the sequences were built on. The criterion is prefix-optimal by construction—each arm’s sequence is the best its bank can do at every budget under the same one-step rule—and that is the property an interview needs, because a question budget is a deployment choice rather than a constant. It is also the property that makes the sixteen-question column the one place a bank can be behind: a prefix-optimal path commits its opening slots on immediate gain and does not revisit them, and NDCG over a sequence is not guaranteed submodular, so containment of the item bank inside the combined bank binds at the optimum rather than along the path. Read across the table, the shape is consistent and it is the shape the rest of this section explains: the concept channel owns the opening and the tail, the item channel owns full accuracy once enough questions have been answered—and how many that is depends on how many the user can answer at all.

The concept channelś advantage is answerability, which is neither new nor surprising. That a cold user can answer a question about a broad property more readily than one about a particular title is close to common sense: the concept question asks whether they like a kind of film, the item question presupposes they have seen a specific one. We claim no discovery here, and the direction is established independently of us. Whether a cold user *can* answer at all is a first-class concern of the classical interview literature, under other names. Rashid et al. select seed items by popularity precisely because popularity is the probability that a new user has seen the item [36], and report that a pure-entropy strategy performs badly for exactly this reason; Golbandi et al.'s tree splits users three ways, into lovers, haters and *unknowns*, so unanswerability is a branch of the method rather than an afterthought [14]; and the active-learning survey of Elahi et al. gives the property its own taxonomy category, as *rateability* or acquisition probability [11]. We adopt the term *answerability* for readability, but we are renaming a known quantity, not introducing one.

The sharpest prior statement of the problem is Sun et al. [43], by three of Functional MF's own authors, whose entire contribution is an answerability fix: they report that a cold-start interview tree becomes so unbalanced that *the unknown branch captures more than* 80% *of users at each split*, so that users must repeatedly answer "unknown" before reaching a question they can engage with, and may simply abandon the interview. Their remedy is to show several items per screen, which raises the chance that a user knows at least one, and they evaluate against interview *time* rather than question count. That is the same pathology we measure, diagnosed twelve years earlier and more sharply. Two things separate our result from theirs. Their fix stays inside the item channel—it makes item questions more likely to land, rather than asking whether a different *kind* of question lands more often—and their multi-question screen is simultaneously more answerable *and* more informative, a confound they do not separate. Our comparison holds the budget and the interview protocol fixed and varies only the channel.

The *shape* of the curve is not new either. Gharahighehi et al. [13] run item-only against item-plus-genre trees and find that attributes lead early while items overtake later; Xia et al. [45] find attributes dominating the opening turns of a dialogue and items the later ones. Both report the crossover we report.

What we do claim, and how far it reaches. Not that concepts are more answerable—the literature above establishes that—but that on this instrument the quantity is *visible*, so a channel comparison can be read against it rather than inferred. In our harness, at an eight-question budget, a user answers 2.18 of the item questions and 6.02 of the concept questions. That figure is a property of *this* setup and we make no claim beyond it: it depends on the question banks (a popularity-ordered item bank is answered 3.0 times in eight, an NDCG-greedy one 2.18), on the structural rule that decides when an item or concept counts as answerable, and on a catalogue where items are films. What generalises is the ordering and its cause, both of which predate us; what is ours is that the instrument reports the number at all, and that the item channel turns out to face a trade-off the concept channel does not—an item bank can be answerable or informative, and the two orderings differ.

On the adaptive headroom. A one-level adaptive probe on the fixed stack—branch on the first answer, then choose the second question per branch, selected and evaluated out-of-sample on disjoint halves—pays on the item channel with a CI excluding zero (+0.0015 full, CI $[+0.0007, +0.0024]$; +0.0016 tail, CI $[+0.0008, +0.0024]$), and is positive but not significant on the concept channel at one level. The effect is small because a one-level preview is not a policy; that is the point. Establishing that adaptivity pays at all, on an instrument whose ingestion is certified, is what this chapter owes the companion method chapter, whose job is to close the gap—the decision-tree acquisition line of Golbandi et al. [14] on *this* instrument and harness. The instrument chapter's job is to make the gap *measurable*; closing it is the method chapter's.

Training concepts into the tower preserves item accuracy but yields no concept gain. We evaluated two homes for the signed concept fold: a trained fold-to-point on the *frozen* tower (call it the frozen-fold), and concept tokens trained *into* the tower itself (the tokens-in-tower variant). Both preserve full-profile strength, as V1 demands. The frozen-fold inherits the tower's V1 by construction (a zero-initialised fold is bit-identical on the full profile). The tokens-in-tower variant is retrained and therefore must re-earn V1, and it does: full/tail NDCG@10 0.3487/0.2471 on the test users, a paired-bootstrap tie with the 0.3540/0.2497 RecVAE bar (difference -0.0053, within the 0.0070 CI), with cold-start accuracy if anything marginally higher. So *training concepts into the tower costs essentially nothing on items.* It also, however, *gains* essentially nothing on concepts, and it is worse where it matters most. On per-answer concept curves the frozen-fold is higher at eight answers (0.1630/0.0639 versus 0.1566/0.0577), and in deployment currency the difference is decisive: the frozen-fold stays above the intercept through sixteen questions (0.1379 at q=16 in the ledger's paired sweep), whereas the tokens-in-tower variant, measured in the same sweep, *craters* to 0.1052, below the no-answer intercept, as the long tail of the fixed bank floods it with weak-support concepts. We therefore carry the frozen-fold as the concept operator of record; the tokens-in-tower arm is item-safe but does not earn its place. The V1 comparison above is a genuine tie on the certified ruler.

The methodological consequence for verification design. The concept channel's near-miss is the sharpest argument in this chapter for *why verification has the shape it does*. A capability check (V5's sign-flip) passed on a fold that, as actually driven down a realistic question bank, never emitted a negative answer—the answer model, not the operator, was sign-blind, and the operator's certified capability was vacuous in deployment. Only the deployment-currency evaluation—a fixed, user-independent bank under a per-question budget, scored at the endpoint (V4's fixed-bank setting)—exposed it, because there the declining curve penalises an answer model that never exercises the negative half. This is a general failure class, not a concept-specific accident: *a gate on what an operator* can *do must be paired with a gate on what the deployed system* does. We record the deployment-currency endpoint as the concept channel's acceptance form, and fold the signed-retrain numbers into the V4/V5 rows of Table 8.

7 Experimental design

7.1 Baseline bank

The baseline bank (Table 6) is three tiers: classic floors that every curve must clear (Dacrema et al. [12] demands them), the full-profile SOTA bar that decides V1, and the elicitation-compatible rivals that decide V3/V4. Reported anchors are on the systems' *own* splits and are not comparable to ours until bridged (§7.2); we print them only to show what "ties SOTA" will mean.

7.2 Evaluation protocol

The protocol items below instantiate the certification bridge and the two outcome checks V1 and V4. The remaining verification items and the two integrity items, which close the false positives those outcome checks admit, are specified in Table 1, and each headline result is reported against them.

(i) The certification bridge. Our ML-25M ruler and the published bar are on *different* metrics and splits, and we do not conflate them. Concretely, our paper-config RecVAE realisation reaches full/tail NDCG@10 0.3540/0.2497 on the canonical ML-25M split—whereas the published EASE/RecVAE numbers are NDCG@100 on the ML-20M strong-generalisation split. These are not comparable, and no "ties SOTA" claim is made until the bridge is run. The bridge has two steps: (1) *reproduce* the published EASE and Mult-VAE/RecVAE numbers on the ML-20M Liang split (NDCG@100, Recall@20/50) in our own code, snapping to the reported values exactly [40, 25, 38]; (2) *port* the certified tower to the ML-25M harness and report full/tail NDCG@10 there. We keep ML-20M in the loop despite being older because the Liang 2018 split is a protocol lock-in the whole full-profile literature reports on; ML-25M is the newer corpus and, crucially, carries the genome tags the concept channel needs, so both datasets are required—one for comparability, one for the channels.

The first step is complete. On the ML-20M Liang strong-generalisation split our split statistics reproduce the published counts exactly (9,990,682 events / 116,677 train users / 20,108 items), and both anchor implementations **snap to their published values**: EASE at NDCG@100 0.4203 vs. the reported 0.420 [40], and RecVAE at 0.4425 vs. the reported 0.442, with Recall@20 (0.4144 vs. 0.414) and Recall@50 (0.5525 vs. 0.553) matching as well [38]. One closed-form and one neural reproduction to the third decimal certify that our split construction, metric implementation, and both implementation families are faithful to the published protocol—every ML-25M baseline number below therefore comes from certified code.

Mult-VAE reaches NDCG@100 0.4223 against the published 0.426 [25] (−0.0037; Recall@20 0.3928 vs. 0.395, Recall@50 0.5341 vs. 0.537)—a *near*-snap, close enough to admit as a bridged Tier-2 anchor but not a third-decimal reproduction, and we label it that way rather than round it into agreement. **Mult-DAE** *fails*: 0.3969 against the published 0.419 (−0.022, with Recall@20 and @50 short by a comparable margin). We do not know whether the gap is a training-recipe detail or something deeper, and rather than carry an uncertified implementation we **drop Mult-DAE** from the baseline bank entirely—it earns no row on the ML-25M ruler.

Consistency demands we apply that rule to ourselves, and doing so costs us something. Our iALS lands at 0.358 against the published WMF anchor of 0.386—a miss of 0.028, *larger* than the 0.022 that disqualified Mult-DAE. It would be a double standard to drop one and keep the other merely because the kept row is convenient, so we state the rule and its consequence explicitly: **iALS is an uncertified floor, not a bridged anchor**. It keeps its measured row on the ML-25M ruler, where it is one more number from the same harness, but it is excluded from every claim that requires a certified comparator—no argument in this chapter rests on an unbridged number being compared with another unbridged number. Only EASE, RecVAE and (as a near-snap) Mult-VAE carry certified status on this ruler. The bifurcation argument must stand on those or not at all.

(ii) V1 — full-profile on ML-25M, items-only. On the baselines' native input (items only), evaluate full and tail NDCG@10 on the full ML-25M catalogue with a paired per-user bootstrap CI, confirming the instrument's belief mean reproduces the frozen tower and ties EASE/RecVAE on the shared harness. The item-only baselines are measured (Table 7).

A properly-trained RecVAE leads the closed-form frontier, and sets the bar. On our canonical ML-25M ruler our in-house paper-config RecVAE attains 0.3540 full / 0.2497 tail, the nominally strongest measured system—ahead of the closed-form leaders EASE (0.3476/0.2441) and EDLAE (0.3433/0.2395). The top of this ruler is a narrow band rather than a ranking: RecVAE leads EASE by +0.0064 full, about 1.8 unpaired standard errors, which is the same order as the 0.0069 paired interval that makes the tower a tie with RecVAE below. We therefore treat RecVAE as *the bar*—the strongest measured system, and the conservative one to be held to—rather than as a system separated from EASE. All rows share the same split, cohort, tail definition, and NDCG@10 routine (verified by code inspection). The point of interest for the instrument is that on this ruler a properly-trained amortised VAE—not a closed-form model—sits at the top of the frontier; the instrument's R1 obligation is precisely to *tie this bar*, reproducing RecVAE-class full/tail accuracy through the frozen tower and then preserving it under the belief layer at V1. We report the whole frontier rather than excluding the stronger baselines.

V1 is passed: the interview tower ties the bar, and is exactly the frozen decoder when the interview is empty. One certified training run of the graded-native set-encoder tower T2′—a single run of the pre-registered winning configuration, selected on validation and scored once on the held-out test cohort—reaches full/tail NDCG@10 **0.3482/0.2462** against the RecVAE bar's 0.3540/0.2497. The difference is -0.0057 full with a paired per-user bootstrap CI of 0.0069, so the two *tie* in the sense of Definition 1: the tower pays no measurable full-profile price for being interviewable. Two properties of that number matter more than its size. First, it is on the *same* ruler as every other row of Table 7—same split, cohort, tail mask, and NDCG routine—so it is a tie in the strict sense, not a bridged one. Second, the V1 *identity* half holds exactly and not merely statistically: with an empty evidence set the tower emits a zero latent ($|\boldsymbol{\mu}_0| = 0$ to numerical precision) and its scores are a rank-identical image of the frozen decoder's learned bias (Spearman 1.0000), which is what licenses reading every later interview curve as evidence about *ingestion* rather than about a shifted baseline. The cold intercept of every curve in this chapter, 0.1279/0.0192, is that same non-personalised read-out. Accuracy is therefore settled for this instrument: V1 is evaluated on the belief mean, so the tie is one the belief layer preserves rather than one it is measured beside, and the layer passes its own gates as well (§7.3). The rankings reported in this chapter are read from the posterior *mean*; putting the covariance to work in question selection is the companion chapter's subject.

7.3 The battery, measured

The battery is proposed in §2 as a portable certificate, so it owes its own results. Table 8 reports every gate.

Within an answerable bank, question choice matters far less than answerability itself. In the V4 curves, a *random* question drawn from the answerable bank beats popularity-ranked items at q=1, 2, 4, 8 (e.g. 0.2358 vs 0.2230 full at q=8) and is only overtaken at q=16 (+0.0022, CI-clean). HELF and the $\boldsymbol{\Sigma}$-greedy selector are likewise within noise of random until the largest budget, where $\boldsymbol{\Sigma}$-greedy wins by +0.0036 (CI $[0.0016, 0.0057]$). This is not a contradiction of V4's fixed-bank arm, whose random arm is drawn from the *whole catalogue* and is therefore mostly unanswerable (mean 1.4 of 16 questions answered, versus 13.5 for the popularity bank) and which the popularity bank beats by +0.1264: the two "random" controls are different objects, and we name them separately. The lesson is that once a bank is restricted to answerable questions, *which* of them is asked matters far less than that they are answerable—which is precisely why the selection problem is deferred to the companion chapter and why we decline to claim a selection result here.

(iii) V4 — per-question curves. Cold-start per-question NDCG@10 curves (full and tail) versus a random-question control. Items are foldable by all baselines; mixed channels (concepts, signed values, continuous tokens) are foldable only by our instrument, which is the point of the comparison—the rivals cap out at what an item-only interview can carry. Symmetric access is enforced where a baseline *can* fold a channel; where it structurally cannot, that is reported as an architecture limit, not hidden. The items-only V4 curves and the V3 shrinkage diagnostic are measured (§7.3), and the mixed-channel comparison is reported in §6.4 under one strategy-free criterion. No item-only baseline appears in that comparison, because none can fold a concept at all—which is the architecture limit itself, argued in (iv) below.

(iv) Why "just add genre columns" is not an alternative. The obvious objection to the concept channel is that a concept could simply be appended to an item-only model as an extra column—a centroid-of-members pseudo-item in EASE, or an extra input unit in RecVAE. The objection fails for a structural reason rather than an empirical one, and the reason is worth stating precisely because it is the same property that motivates the whole channel.

A column is fixed when the model is fitted. EASE's item–item weight matrix is defined over a fixed column set and is obtained in closed form from the Gram matrix of exactly those columns; RecVAE's input layer has one weight vector per column. Adding a concept therefore means adding a column, and adding a column means *refitting the model.* This is not a tuning inconvenience: it means the set of expressible concepts is frozen at training time. A paraphrase of an existing concept, a concept invented after deployment, a direction produced by snapping a continuous query to the nearest meaningful axis—none of these can be answered without retraining, because none of them has a column. The pseudo-item is not merely a lossy encoding of a concept; it is a *closed* one.

Metrics. Full *and* tail NDCG@10 on ML-25M (the project reporting rule, scored with the decoder's learned bias, not a log-count popularity floor), plus NDCG@100 (and Recall@20/50) on the ML-20M bridge split for comparability with the published anchors. Every headline delta carries a paired per-user bootstrap CI on one shared harness. Every cell reported is measured on this harness.

8 Limitations and threats to validity

- **Metric non-comparability until the bridge is run.** Our frontier number 0.3540/0.2497 is NDCG@10 on ML-25M; the published bar is NDCG@100 on ML-20M. Until §7.2 snaps our code to the published numbers (done for EASE/RecVAE, §7.2), *no* "ties SOTA" claim across the two rulers is licensed—the ML-25M and ML-20M numbers live on different rulers.
- **Monotonicity is empirical, not a theorem.** Conjugate-Gaussian updates are monotone in expected utility [15, 6] *only in expectation*, and only if the ranker acts optimally under the belief—which a fixed dot-product / NDCG ranker does not. R4 (and V4's rising curve) is therefore a *measured* property with a per-question CI, not a guarantee; a truthful answer can in principle still lower NDCG at a given step, and we report any such step rather than suppress it.
- **Answers come from recorded behaviour.** V3 and V4 are measured under the answer model that integrity item I2 fixes: an item answer is the user's own recorded rating and a concept answer is a log watch-lift over raw counts and item marginals, so no recommender-derived quantity can enter an answer and the self-simulator inversion is closed by construction rather than by measurement. These are real preferences, but they are not a person responding

to a rendered question, and the chapter makes no claim about how a user would phrase, hedge or misread one.

- **BERT4Rec replicability.** The sequence baseline's fold-in advantage is subject to a documented replicability caveat [34]; we tune it to the published protocol and flag any residual gap rather than treat a single run as authoritative.
- **Graph-filter admissibility.** GF-CF/BSPM/Turbo-CF [37, 8, 32] do not publish an ML-20M strong-generalisation NDCG, so no row in this family can be a *bridged* tie. Turbo-CF is re-run on our split with ($\alpha, s,$ filter) selected on our validation users and is reported as best-effort tuned; GF-CF and BSPM are not run here and carry no number.
- **Scope of the novelty claim.** The conjunction claim is bounded by our survey: a set-encoder / deep-set recommender that reached RecVAE-class full-profile accuracy *and* took value-carrying mixed-type tokens would partly pre-empt R1∧R2∧R3, and we found no such reported system.

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Family	Representatives	R1	R2	R3	R4	R5	One-line verdict
(i) retrain / meta-learn per user	MeLU, TaNP [26], CDRNP	✗	✗	✓	✗	✓	Set-input and belief precedent; not full-profile SOTA, not channel-open.
(ii) fold-in linear / closed-form	EASE [40], SLIM, GF-CF [37], BSPM [8], Turbo-CF [32]	✓	✗	✗	✗	✗	EASE/SLIM hold the bar (V1); the graph filters do not on this ruler. Item-indicator basis $\Rightarrow$ R2, R5 impossible.
(iii) amortised inference encoders	Mult-VAE [25], RecVAE [38], EDLAE [41], EDDI [30]	✓	✗	✗	✗	✗	The architecture we live in; VAEs item-only, EDDI right shape/wrong domain.
(iv) Bayesian belief over item space	Bıyık [5], ConTS [23], BCIE [44], ICF [46]	✗	✗	✓	✗	✓	The belief heart; belief layers, not full-profile towers.
(v) sequence models	SASRec [20], BERT4Rec [42]	✗	✗	✗	✗	✗	Interview $\neq$ ordered session; tokens item-only, order-sensitive.
(vi) LLM-as-recommender / CRS	PEBOL [3], GATE [22], LLM-CRS [17]	✗	✓	✗	✗	✓	Best R2 (open text); weakest R1 (weak collaborative signal).
CASPER (delivered)	**this chapter**	✓	✓	✓	✓	✓	**The conjunction; no published system occupies it.**

Table 2: Six-family taxonomy scored against R1–R5. ✓ met, ✗ not met—there are no half marks: a mark is met only if the system provides the capability as a designed, demonstrated feature. Scores for the six published families are our reading of their *published* claims and are qualitative; the last row is this chapter's instrument, whose marks are the measured verification results of Table 8 rather than a reading. R1 and {R2,R3,R4,R5} are anti-correlated across the published families: (ii)/(iii) hold the accuracy corner, (iv) the belief corner, and none spans both.

Most-Popular	✗	✗	✗	✗	✗	.1345 / .0226	measured	n/r (non-personalised floor)
Item-kNN (cosine)	✗	✗	✗	✗	✗	.2428 / .1297	measured	@100 ∼ 0.36 class [9]
PureSVD / iALS-WMF	✗	✗	✗	✗	✗	.2442 / .1853	measured (iALS)	@100 0.386 (WMF, ML-20M)
EASE [40]	✓	✗	✗	✗	✗	.3476 / .2441	measured	@100 0.420 (ours **PASS** +.0003)
EDLAE [41]	✓	✗	✗	✗	✗	.3433 / .2395	measured	@100 ≈ 0.42–0.44 class
Turbo-CF [32]	✗	✗	✗	✗	✗	.2622 / .1621	measured (val-tuned)	n/r (ML-20M not reported)
GF-CF / BSPM [37, 8]	✗	✗	✗	✗	✗	—	not run here	n/r (ML-20M not reported)
(b) Amortized / neural full-profile								
Mult-DAE [25]	✓	✗	✗	✗	✗	—	*dropped* (snap failed)	@100 0.419 (ours 0.397, **FAIL** −.022)
Mult-VAE [25]	✓	✗	✗	✗	✗	.3202 / .2194	measured	@100 0.426 (ours 0.422, near-snap −.004)
RecVAE [38] (paper config; our run)	✓	✗	✗	✗	✗	**.3540** / **.2497**	measured	@100 0.442 (impl. **PASS** +.0005)
SASRec [20]	✗	✗	✗	✗	✗	.1486 / .0805	measured (best-effort)	n/r (next-item task)
BERT4Rec [42]	✗	✗	✗	✗	✗	—	not run here	n/r (repl. [34])
TaNP [26] (MeLU / CDRNP line)	✗	✗	✓	✗	✓	.2645 / .1635	measured (best-effort)	n/r (own cold-start benchmark)
(c) Elicitation / CRS systems — the belief corner								
EDDI / Partial-VAE [30]	✗	✗	✓	✗	✓	—	not run here	n/r (non-recsys; UCI/MNIST)
Golbandi tree [14]	✗	✗	✗	✗	✓	.3065 / .1895	measured (core)	n/r (Netflix RMSE)
RBMF / Functional-MF [27, 47]	✗	✗	✗	✗	✓	.2534 / .1764	measured (best-effort)	n/r (elicitation RMSE)
Bıyık soft-attributes [5]	✗	✗	✓	✗	✓	—	reports no full-catalogue accuracy	n/r (16 users, slates of 5)
ICF [46]	✗	✗	✓	✗	✓	—	core ≈ iALS (measured)	n/r (online CF; own protocol)
ConTS [23]	✗	✗	✓	✗	✗	—	reports no full-catalogue accuracy	n/r (cold-start CRS metrics)
BCIE [44]	✗	✗	✓	✓	✗	—	not portable (needs their KG)	n/r (critique metrics)
UNICORN [10]	✗	✗	✓	✗	✗	—	not portable (needs their KG)	n/r (graph-RL CRS; own metrics)
PEBOL [3]	✗	✓	✓	✗	✗	—	not runnable at full catalogue	MRR@10 0.27 vs 0.17 (cold-start only)
GATE [22]	✗	✓	✗	✗	✓	—	not runnable (no ranker)	n/r (elicitation-only; no full ranking)
LLM zero-shot CRS [17]	✗	✓	✗	✗	✓	—	not runnable at scale	n/r (weak collaborative signal)
(d) Ours								
The instrument (this chapter)	✓	✓	✓	✓	✓	**.3482** / **.2462**	**measured (certified)**	n/a (in-house; graded-native tower)

Table 3: **Master table: every reasonable system by R1–R5 and by measured full-profile accuracy on one shared ruler.** ✓ met, ✗ not met, with no intermediate grade: a requirement counts as met only where the system provides it as a designed, demonstrated capability (our reading of each system's *published* claims, consistent with Table 2). *full/tail NDCG@10* is on the canonical Liang-recipe strong-generalization ML-25M split (catalog restricted to the 18,430-item project catalog, train vocabulary 18,359; train 140,768 / val 10k / test 10k users; per-user 80/20 fold-in; tail = top-33%-train-mass head mask, 190 head items), scored with the decoder's learned bias. The Bıyık and ConTS rows carry a dash because neither reports full-catalogue accuracy; the Golbandi row prints its node-recommender core. Statuses are *measured* (run on this ruler; best-effort = with stated substitutions), *not run here* (runnable in principle, no number claimed), or *not runnable / not portable* (impossible by construction; mechanism in the text). The final column is each system's own published full-profile anchor on its own split/metric—*not* comparable to our ruler until bridged (§7.2); "n/r" where the literature reports none. The RecVAE row is our own paper-config training run on this ruler and is the current full-profile frontier; the "ours" block adds

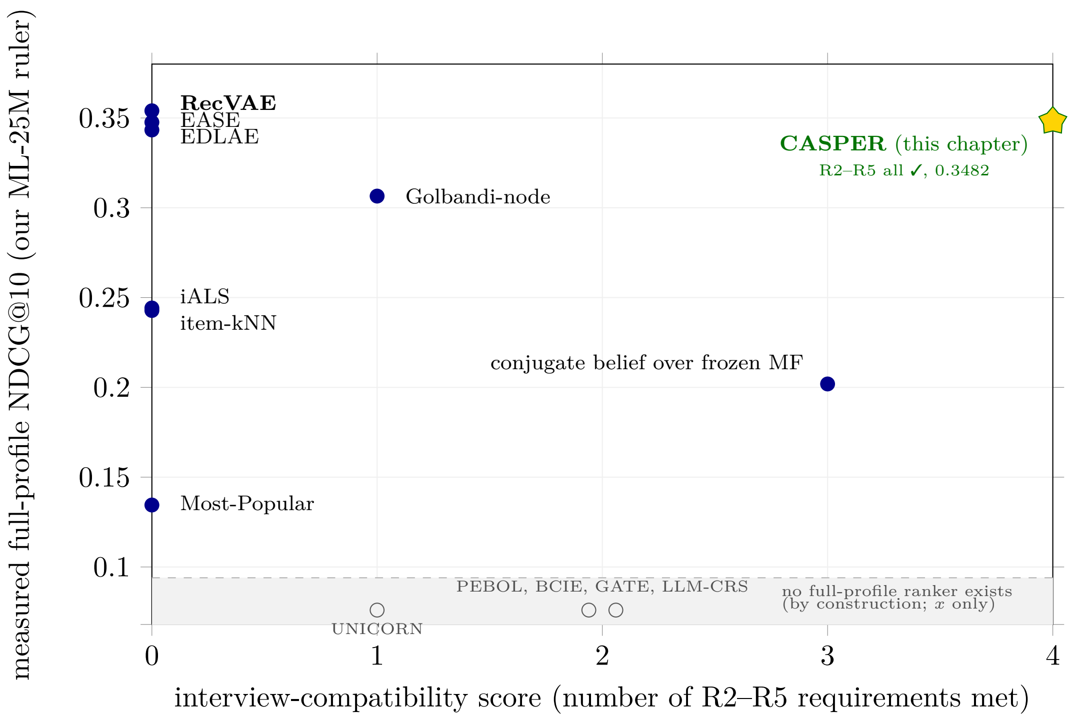

Figure 1: **The gap map (measured axes).** y-axis: measured full-profile NDCG@10 on our canonical ML-25M ruler. x-axis: an *interview-compatibility score*, defined as the number of **R2–R5** marks a system carries in the master table (Table 3), maximum 4. There are no half marks, so the axis is an integer count. *R1 is deliberately excluded from the score*: accuracy is the y-axis, and letting it also inflate x would blur the very trade-off the figure exists to show. The score is a *coarse ordinal capability count, not a measurement*, so the x-axis is qualitative even though the y-axis is measured. Filled markers are systems with a full-profile number on this ruler; Golbandi-node is that tree's best-effort node recommender, and the conjugate-belief point is our own posterior-mean ranker (§5), plotted to show what ranking on a conjugate mean costs. Hollow markers in the separated grey strip *below* the measured y-scale are systems that *cannot produce a full-profile number by construction* (no full-catalogue ranker, or not portable; see the master-table dispositions); only their x position is meaningful. Runnable-but-not-yet-measured systems (sequence models, graph filters, EDDI, TaNP) are *absent* from the figure and join the plot when their numbers land. The star is this chapter's certified instrument at 0.3482—the only measured point that is simultaneously at the frontier bar and at full interview compatibility. The instrument sits at x=4—no published system exceeds two—the only point in the top-right cell: R2 by V2's member lift and its open-vocabulary fold, R3 by V3's directional and load-bearing covariance, R4 by V4's rising fixed-bank curve, and R5 by V5's sign and intensity arms—each verified rather than asserted (Table 8). The population is otherwise *bimodal*: the entire accuracy corner (Pop through RecVAE, 0.343–0.354 at the top) collapses onto x=0—strong recommenders that can fold items and nothing else—while every system with genuine interview machinery sits at $x \geq 1$ and reports no full-catalogue accuracy at all. The one point plotted at x=3 is not a rival but our own conjugate posterior-mean ranker at 0.2019, included because it shows what the uniform-fold alternative costs. Two clusters, no bridge: the corner is not expensive; it is unoccupied. The nearest published rival to the target is Bıyık et al. [5], the highest compatibility of any system with an actionable belief.

free-text phrase	nearest concept	cos	top of the returned ranking
“the quiet dignity of losing”	*redemption*	0.47	Umberto D.; Tyson; Red Army; Days of Glory
“people trapped by a decision they made years ago”	*history*	0.39	Music Box; Judgment at Nuremberg; Missing
“two people talking for the whole film”	*dialogue*	0.54	Sweet Smell of Success; Trouble in Paradise
“movies about grief”	*feel good movie*	0.54	Rachel, Rachel; Mean Creek; The Woodsman

Table 4: **A phrase with no curated counterpart is placed, not looked up.** Four free-text phrases, each screened against the 1,031-concept vocabulary two ways—no genome tag occurs as a whole-word substring, and no concept sits within cosine 0.6—with the nearest concept we actually hold printed so the distance from anything curated is visible. Directions come from the single global map fitted on 80% of the concepts; nothing here is retrained or looked up.

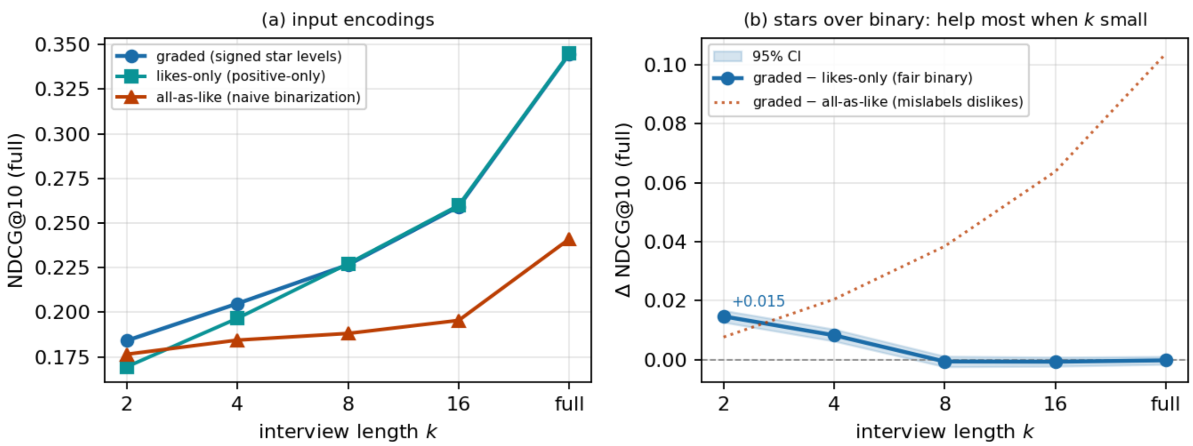

Figure 2: **Graded input helps over binarized input, most on the short interview** (ML-25M interview harness, full NDCG@10 vs. interview length k). **(a)** The same revealed set is fed three ways: *graded* (real signed star levels), *likes-only* (sub-threshold answers dropped—exactly what a positive-only tower such as RecVAE does at $r>3.5$), and *all-as-like* (every answered item recorded as a 4-star like—naive implicit binarization). **(b)** Against the fair positive-only baseline, the graded premium is *CI-clean positive on the short interview*—+0.0147 at $k=2$ (CI $[+0.0127, +0.0167]$), $+0.0083$ at $k=4$ (CI $[+0.0063, +0.0103]$)—and fades to a tie from $k=8$ onward (-0.0006, CI $[-0.0024, +0.0012]$) as collaborative signal fills in: the extra bits a star carries are most decisive when few items are known. The *mechanism* is that a signed answer encodes a dislike (a negative y_j the item-indicator basis cannot express); against the *all-as-like* control, which mislabels dislikes as likes, the apparent premium instead *grows* with k (dotted)—a control artifact (more revealed items $\Rightarrow$ more dislikes corrupted), not value from gradation. Mirroring the star signs costs -0.2985 on the full-profile arm of V5 (Table 8).

bank	q=1	q=2	q=4	q=8	q=16	answered of 8
items-only	.1405/.0275	.1524/.0387	.1656/.0445	**.1863**/.0538	**.2098**/.0679	2.18
concepts-only	**.1436**/**.0323**	.1504/**.0434**	.1647/**.0551**	.1776/**.0698**	.1808/.0758	**6.02**
combined	**.1436**/**.0323**	.1504/**.0434**	.1647/**.0551**	.1817/.0616	.2049/**.0776**	4.67

Table 5: **Best static question sequences, one strategy-free criterion per bank.** Full/tail NDCG@10 at each budget, sequences built by lazy greedy on the 10,000 validation users over a 500-item and/or 500-concept bank and scored once on the 10,000 disjoint test users; no refusals; intercept 0.1279/0.0192. The last column is the mean number of the first eight questions a user can actually answer. Bold marks the best bank in each column. For comparisons of this kind—two *arms of the same model* over the same users, whose per-user differences are strongly correlated—paired bootstrap intervals on 10,000 users have half-widths of ±0.002–0.003, and the differences read in the text are the ones comfortably above that. Intervals for comparisons *across models* are wider, because two systems' per-user scores are far less correlated: the tower-versus-RecVAE interval of §7 is 0.0069.

Baseline	Type	Reported anchor	Why the slot
Tier 1 — classic floors (must clear)			
Most-Popular	non-personalised	popb floor (our harness)	the "did we learn anything" line; the V4 curve must clear it at budget.
Item-kNN (cosine)	neighbourhood	ML-20M NDCG@100 ~ 0.36 class	instant fold-in; Cremonesi-era strong-simple [9].
PureSVD / iALS-WMF	linear MF fold-in	WMF ML-20M NDCG@100 0.386	cheap fold-in reference.
SLIM	sparse linear item-item	ML-20M NDCG@100 0.401	direct EASE ancestor.
Tier 2 — full-profile SOTA (the R1 / V1 bar)			
EASE [40]	closed-form item-item	NDCG@100 0.420	the bar to tie; item-only shows the R3 gap.
RecVAE [38]	amortised VAE	NDCG@100 0.442	top of the ML-20M split; our in-house frontier.
Mult-VAE [25]	amortised VAE	NDCG@100 0.426 (ours 0.422)	canonical amortised anchor; near-snap.
Mult-DAE (dropped)	DAE	NDCG@100 0.419 (ours 0.397)	snap **failed** -0.022; disqualified.
EDLAE [41]	denoising higher-order EASE	$\approx$ RecVAE class	confirms the flat frontier.
GF-CF / BSPM / Turbo-CF [37, 8, 32]	closed-form graph filter	SOTA on Gowalla/Yelp (ML-20M not reported)	admit only if re-run on our split, which Turbo-CF now is: 0.2622/0.1621, val-tuned.
SASRec / BERT4Rec [20, 42]	self-attentive sequence	strong next-item (not this split)	sequence fold-in; SASRec measured at 0.1486/0.0805, far off the bar.
Tier 3 — elicitation-compatible rivals (decide V3/V4)			
EDDI / Partial-VAE [30]	set-encoder + info-gain	non-recsys (UCI/MNIST)	the R2 architecture; myopic info-gain baseline.
Bıyık soft-attributes [5]	Gaussian belief + EVOI	own elicitation task (16 users, slates of 5)	the central threat; open-vocabulary and R1-bar wedge.
ConTS [23]	Thompson bandit, item+attr arms	cold-start CRS metrics	unified-space + uncertainty; fixed schema.
BCIE [44]	conjugate-Gaussian critique fold	critique metrics	exact-conjugacy-over-frozen precedent.
PEBOL [3]	Bayesian-opt NL-PE (LLM)	MRR@10 0.27 vs 0.17	load-bearing PE baseline; per-item Beta.
Golbandi tree [14]	RMSE decision-tree interview	Netflix RMSE	the static-tree interview comparator.
RBMF / Functional-MF [27, 47]	MF-embedding seeds	elicitation RMSE	interview-seed line; cite-and-differentiate.

Table 6: Baseline bank. Anchors are on each system's own split/metric and are *not* comparable to ours until the bridge (§7.2) is run; V1 is decided against EASE+RecVAE, V3/V4 against the Tier-3 rivals plus a random-question control. Any deep-net "SOTA" not snapping to Tier-2 numbers on our split is inadmissible [12].

Method	full NDCG@10	tail NDCG@10	note
Most-Popular	0.1345	0.0226	@100 0.1975; non-personalised floor
SASRec [20]	0.1486	0.0805	@100 0.2465; sequence fold-in, best-effort
conjugate belief over frozen MF	0.2019	0.1200	@100 0.2886; the posterior-mean ranker (§5)
Item-kNN (cosine)	0.2428	0.1297	@100 0.3200; neighbourhood fold-in
iALS-WMF	0.2442	0.1853	@100 0.3704; MF fold-in
RBMF / Functional-MF seed	0.2534	0.1764	@100 0.3607; best-effort seed core
TaNP [26]	0.2645	0.1635	@100 0.3661; best-effort, rating channel binarised
Turbo-CF [32]	0.2622	0.1621	@100 0.3569; val-tuned α=0.5, s=1, 2nd-order
Golbandi node recommender	0.3065	0.1895	@100 0.3958; best-effort tree core
Mult-VAE [25]	0.3202	0.2194	@100 0.4303; near-snap -0.004
EDLAE [41]	0.3433	0.2395	@100 0.4330; val p=0.4
EASE [40]	0.3476	0.2441	@100 0.4375
RecVAE (paper config; our run) [38]	**0.3540**	**0.2497**	@100 0.4537; nominal frontier
Ours			
Set-encoder tower T2′ (certified)	0.3482	0.2462	@100 0.4486; **V1 tie** (-0.0057, CI 0.0069)

Table 7: V1 — full-profile, items-only, on the canonical Liang-recipe strong-generalization ML-25M split (catalog restricted to the 18,430-item project catalog, train vocabulary 18,359; train 140,768 / val 10k / test 10k users; per-user 80/20 fold-in; seed 98765), NDCG@10 (full and tail); tail = top-33%-train-mass head mask (190 head items); scored with the decoder's learned bias. The belief-MF, RBMF and Golbandi rows are best-effort cores of the belief-corner systems (§3). All rows reuse the canonical harness verbatim, so they are on one ruler.

Check	Criterion	Measured	Verdict
V1	tie the bar; identity at t=0	0.3482/0.2462 vs 0.3540/0.2497, diff -0.0057 (CI 0.0069); $\lvert\boldsymbol{\mu}_0\rvert$=0, Spearman 1.0000 vs decoder bias	**PASS**
V2	open set-valued input: any t, and directions not fixed at fit time	set half by construction (the fold is a running sum, t=0 is the prior). Open half measured: the direction ranks its members at AUC 0.94 whitened (raw 0.44, below chance—whitening is what makes a concept a direction); the fold is *concept-specific*—a like lifts c's members to AUC 0.626, a dislike pushes them to 0.416, and folding a *different* answerable concept leaves them at chance (0.498), giving a sign contrast of +0.2096 (CI $[+0.1986, +0.2208]$) and an identity contrast of +0.1278 (CI $[+0.1195, +0.1361]$); it beats the concept's own popularity projection by +0.0087; and a concept *held out of the adapter's fit* folds from its name alone and recovers its own top-10 at $> 300\times$ chance	**PASS**
V3	$\boldsymbol{\Sigma}$ shrinks, is directional, and is load-bearing	tr $\boldsymbol{\Sigma}$ 1881.9 $\to$ 1805.1 monotone; along-$\boldsymbol{\phi}$ 0.996 vs 0.086 elsewhere = **11.5**$\times$; beats isotropic $c\mathbf{I}$ by +0.0089 at q=16 (CI $[0.0071, 0.0107]$); order deviation 5×10^{-7}	**PASS**
V4‡	cold curve rises on a bank fixed in advance	0.1279 $\to$ 0.2724 full, 0.0192 $\to$ 0.1305 tail; fixed-bank rise +0.1218 (CI $[0.1184, 0.1252]$), monotone ρ = 1.0; canary: one answer from cold clears the intercept on both channels (items 0.1405/0.0275, concepts 0.1436/0.0323)	**PASS**‡
V5	sign and intensity are carried	sign-flip 0.4279 $\to$ 0.1294, -0.2985 (CI $[-0.3047, -0.2922]$); neutralisation 0.4190 $\to$ 0.1321, tail 0.2999 $\to$ 0.0368; intensity staircase ρ = 0.927; graded over a fair binarising ablation +0.0147 at k=2 (CI $[+0.0127, +0.0167]$) and +0.0083 at k=4	**PASS**
I1	wrong-user / shuffle / placebo / duplicate gain nothing	true arm beats shuffled by +0.0371 and placebo by +0.0168 (both CI-clean); duplicate assert fired; wrong-user -0.0986 (a stranger's taste is worse than silence)	**PASS**
I2	answers carry no recommender geometry, no target half	item answer = own rating; concept answer = log watch-lift over raw counts, both read from the fold-in only	**by construction**

Table 8: **Verification, measured.** Every requirement is checked and every check passes. V4 gates the covariance on its *structure* and not on the absolute scale of its uncertainty: the scale question is confounded by target difficulty—the mean popularity of a user's held-out items predicts their NLL with $\lvert\rho\rvert = 0.70$, where the belief's own σ reaches 0.24—so a scale-calibration check would largely measure which users happen to have easy targets. The directional margin, 11.5× against a 2× bar, is the load-bearing evidence that the covariance carries per-direction information. ‡V4's cold curve is measured on a bank filtered per user to the questions they can answer, so it isolates *ingestion* from question *availability*; a deployed static sequence faces availability separately (§6). V5's sign and neutralisation arms are *full-profile* corruptions—the user's whole rated history is folded and then flipped or neutralised—so their uncorrupted references (0.4279 and 0.4190 full, 0.2999 tail)